

# Software-as-a-Graph: Benchmarking Centrality and Graph Learning for Pre-Deployment Cascade-Impact Ranking in Publish–Subscribe Systems

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## Abstract

Publish–subscribe middleware hides the paths along which failures cascade, so architects lack pre-deployment evidence on which components are systemically critical. Software-as-a-Graph (SaG) models event-driven architectures as typed multigraphs, derives explicit dependency graphs from deployment manifests, and ranks components by simulated cascade impact. We benchmark closed-form centrality, graph neural networks and hybrid engines on twelve synthetic architectures under leave-one-scenario-out evaluation and on five hand-authored models of open-source systems, against three simulators. The reachability simulator’s first propagation wave is exactly a component’s number of direct dependents, so dependency counts are reported as references that restate the oracle, not as predictors: its first-order expansion reaches Spearman  $\rho = 0.808$  and the direct-dependent count 0.764. The registered primary contrast, a heterogeneous graph transformer against weighted centrality, was null ( $\Delta\rho = +0.069$ ,  $p = 0.27$ ). Hybrids that correct a learned engine with the centrality prior beat centrality on 11 of 12 architectures (+0.103 and +0.130, Holm-significant). Graph attention networks reading the dependency graph reached 0.748, approaching the reference without exceeding it, while also reading an in-degree feature. On a multi-criteria simulator, weighted centrality (0.702) ranked above every learned engine ( $\leq 0.585$ ). On the five system models, learned engines reached about 0.81 zero-shot, against 0.53 for centrality. One reachability simulation took 0.08–4.5 s, less than the neural pipeline’s feature extraction. The result is a benchmark with a largely negative outcome for graph learning: no learned engine exceeds a restatement of the oracle’s first propagation wave, and where that oracle is cheap, running it directly is the simplest choice.

**Keywords:** Dependency graphs, cascading failures, publish–subscribe, graph neural networks, software architecture, dependability, empirical study

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## 1. Introduction

### 1.1. Motivation and Problem

Large distributed systems increasingly communicate through asynchronous publish–subscribe (pub-sub) middleware: ROS 2 in autonomous driving [1], Apache Kafka in enterprise event streams [2], DDS in cyber-physical systems [3], MQTT in IoT [4], and cloud-native microservices [5, 6]. Pub-sub decouples producers and consumers in space, time and synchronization [7]. Components interact through topics and brokers rather than direct references, and deployment-time Quality-of-Service (QoS) policies govern reliability, durability, priority and deadlines.

The same decoupling hides how failures spread. Publishers and subscribers share no direct link, so outages, head-of-line blocking and backpressure propagate along concealed paths through brokers, shared topics, colocated hosts and shared libraries [8, 9]. These failures take two forms. In *sequential cascades*, a slow subscriber fills a broker queue and gradually starves its publishers [10]. In *simultaneous blasts*, a

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shared-library crash or host outage takes down every colocated service at once. Neither architecture diagrams nor static call graphs show these mechanisms. The cheapest time to reduce the risk is before deployment, at design and continuous-integration time [11, 12], when no runtime telemetry exists. Architects therefore need to know, from configuration manifests alone, which components, topics and links are systemically critical and why.

Existing practice leaves this gap open, which we call the **Architecture–Code Gap**: a system can have bug-free code in every service and still be fragile through hidden single points of failure or mismatched QoS contracts [13, 14]. Architecture evaluations such as ATAM rely on manual elicitation [15]. Static code analysis inspects services in isolation [16, 17]. Chaos engineering [18] needs a provisioned cluster, while pre-production alternatives like service-level fault injection testing [19] and microservice dependency tracing [20] require runnable execution environments. Homogeneous centrality flattens typed topologies into untyped graphs [21, 22]. Learned models, graph neural networks in particular, could combine these structural cues. What is missing is a representation that makes pub-sub failure paths explicit, and evidence on which analyzer to trust on it: whether graph learning adds anything over the coupling metrics that architecture research already has. This paper provides that evidence for one well-defined task, ranking components by simulated cascade impact before deployment.

### 1.2. The Software-as-a-Graph (SaG) Approach

**Software-as-a-Graph (SaG)** is a pre-deployment static analysis framework for event-driven architectures (Figure 1). It (1) models an architecture as a typed, directed multigraph over Applications, Brokers, Topics, Execution Hosts and Shared Libraries (Section 3.1); (2) derives from it, through publish–subscribe rules, an explicit `DEPENDS_ON` dependency graph (Section 3.3); and (3) ranks components by predicted cascade impact with training-free rankers, graph-learning engines, and hybrid engines in which a learned model corrects a closed-form score (Section 4). Predictors read only the analysis graph. Ground truth comes from three simulators that run on the raw structural topology (Section 4.3).

**What the ground truth is, and is not.** All ground truth in this study is simulated. The primary oracle  $I^*$  is a reachability simulation that runs on the same manifest the predictors read, and its first propagation wave is exactly the number of a component’s direct dependents (Section 4.4). Dependency counts are therefore reported as references that restate the oracle, never as predictors. The study measures agreement between cheap static rankers and simulators, not the prediction of observed outages. Because ranking against  $I^*$  rewards restating its propagation rule, we bound the circularity with two further simulators and with partial correlations, and report the cost of running  $I^*$  itself (Section 6.4); validation against observed failures remains open (Section 7.3).

**Findings in brief.** The registered primary contrast, a heterogeneous graph transformer against QoS-weighted centrality, was null ( $+0.069$ ,  $p = 0.266$ ). The only confirmatory gains are two hybrids, learned engines corrected by the centrality prior, which beat centrality on 11 of 12 synthetic architectures ( $+0.103$  and  $+0.130$ , Holm-significant). Rankings that restate  $I^*$ ’s propagation rule set a reference level on it: the oracle’s first-order expansion reaches  $\rho = 0.808$  and the number of direct dependents (`InDeg`, publish–subscribe afferent coupling [23, 24])  $0.764$ . In exploratory analyses, graph attention networks on the dependency graph approach that reference ( $0.748$ ) without exceeding it, and they receive an in-degree feature, so they do not escape the circularity either. On a multi-criteria simulator, QoS-weighted centrality ( $0.702$ ) ranks above every learned engine. The result is a benchmark with a largely negative outcome for graph learning on this task.

### 1.3. Research Questions

- **RQ1 (Ranking accuracy):** *How accurately do SaG’s closed-form, learned and hybrid engines rank components by cascading-failure impact on unseen architectures, compared with structural baselines, and how close do they come to reference rankings that restate the oracle’s propagation rule?*
- **RQ2 (What learning needs):** *What do learned engines need: which graph they read, relation-specific (typed) parameters, or QoS inputs, once model capacity and edge-channel width are matched?*
- **RQ3 (Transfer):** *How well do engines trained on synthetic architectures transfer zero-shot to independently authored models inspired by five open-source systems?*

- **RQ4 (Cost):** *What does the analysis cost at CI/CD time, which stage dominates, and how does it compare with running the simulation directly?*

The evaluation follows a registered protocol. Only the primary contrast, the matched  $2 \times 2$  and the two hybrid contrasts are confirmatory; the dependency-count references, the dependency-graph learners, the raw-graph baselines and the cross-oracle analyses were registered in later amendments, after the primary result was known, and are reported as exploratory (Section 5.3). Supplementary Section S25 logs every amendment and maps these four research questions onto the registered plan.

#### 1.4. Contributions

Evaluated under leave-one-scenario-out (LOSO) cross-validation over twelve synthetic architectures and on five hand-authored models of open-source systems, this paper contributes:

1. **A benchmark of static rankers for pre-deployment cascade-impact ranking in publish-subscribe systems** (Sections 5–6): seventeen architectures (2,812 components), three simulation oracles, and training-free, learned and hybrid rankers, with dependency counts and the oracle’s first-order expansion reported as references that restate the oracle (Amendment 13), with a registered analysis plan, its amendments, and artifacts that regenerate byte-identically.
2. **A largely negative result for graph learning on this task** (Sections 6.1–6.2). The registered primary contrast is null; only the two hybrids beat closed-form centrality under correction. Graph neural networks gain +0.08 to +0.11 by reading the dependency graph instead of the raw multigraph, reaching 0.748, but never exceed a count that restates the oracle’s first wave (0.764), and on the raw multigraph they act as per-node models. Relation typing adds nothing at matched capacity (main effect  $-0.014$ ), and in the single untuned configuration tested the heterogeneous transformer is unstable across seeds.
3. **A bounded account of the oracle circularity** (Sections 4.4 and 6.1). Ranking against a reachability oracle rewards restating its propagation rule, which is why the dependency counts are reported as references and not as predictors. The paper measures how far the references reach on  $I^*$ , how much of them survives on a queue-flow simulator beyond  $I^*$  (partial correlation 0.259 [0.143, 0.367] for the direct count, none for transitive reach, on a 30-Application sample per fold), and that on a multi-criteria simulator QoS-weighted centrality ranks above both the references and every learned engine.
4. **Explicit dependency rules for publish-subscribe architectures** (Section 3), which state the typed query behind afferent coupling and add library-mediated blasts.
5. **A cost profile** (Section 6.4): one run of the reachability oracle takes 0.08–4.5 s per corpus architecture, and neural feature extraction costs median  $5.6\times$  that, so where the oracle’s ranking is wanted, running it directly is cheaper than approximating it.

A previous conference paper [25] introduced the publish-subscribe multigraph, the Application-level subscriber→publisher dependency (Rule 1 of Table 3) and a closed-form betweenness-articulation score of the same form as *Topo* (Eq. (5), with weights 0.7/0.3), and validated that score against a reachability-loss simulation on synthetic topologies and two ROS 2 benchmarks. Everything else is new in this paper: the Library and infrastructure rules (Rules 2–6, including library-mediated blasts), QoS weighting, the reference counts, the learned and hybrid engines, the QoS edge encoding, LOSO and zero-shot evaluation, the second and third oracles, the matched control and the cost profile.

Section 2 reviews related work, Sections 3–4 present the model and the rankers, Sections 5–6 the evaluation, Section 7 the discussion and threats, and Section 8 concludes.

## 2. Related Work

### 2.1. Dependability Analysis of Distributed Systems

Runtime approaches to dependability, such as broker clustering, backpressure, autoscaling, failover and chaos engineering [18], require a running cluster, can disrupt service, and consume substantial compute. That

compute is itself a concern of green software engineering [26, 27, 28]. Pre-production alternatives, including service-level fault injection testing [19] and microservice dependency localization via PageRank (MicroRank [20]), provide lighter options before deployment. Benchmark systems with cataloged, replayable faults, such as Train-Ticket [29], make observed failure impact measurable on a running deployment, which is the validation this study does not yet have (Section 7.4). Architecture-based reliability prediction has a long history. Yacoub and Ammar’s methodology for architecture-level reliability risk analysis [30] pioneered combining component dependency graphs with complexity and failure severity to rank components by risk. Cheung’s absorbing Markov chain [31] and the state-, path- and additive models surveyed by Goseva-Popstojanova and Trivedi [32] and Immonen and Niemelä [33] are classic examples, as are the Palladio Component Model [34], layered queueing networks [35] and the AADL Error Model Annex [36]. These approaches answer broader questions but need operational profiles and failure rates that are unavailable at commit time. SaG asks a narrower question from manifests alone: whose failure reaches furthest in the declared topology?

Telemetry-driven methods diagnose faults in running microservices. Examples are Seer [37] and Sage [38], MicroRCA [39], TraceRCA [40], and GNN-based DeepTraLog [41], Eadro [42] and MicroCause [43] (reviewed in [44]). In synchronous microservices, cascades stem from thread-pool exhaustion, RPC timeouts and retry storms [45]. In pub-sub systems they spread through queue saturation and message starvation. All of these methods need traces or metrics from a live system; SaG works before any code runs. SaG’s question is also that of change impact analysis [46] and of dependency management in microservice fleets [47]. Ranking by impact relates to learning-to-rank defect prediction [48], which optimizes the ranking measure directly, as our listwise loss does (Section 4.2).

## 2.2. Static Code Analysis and Static System Analysis

Static code analysis (SCA) tools such as SonarQube [16] measure complexity [49], cohesion and coupling [17, 50] within individual services to flag defect-prone modules [51, 52, 53, 54]. Comparing sophisticated graph measures against simple structural metrics has a rich history in software engineering defect prediction: Zimmermann and Nagappan [55] evaluated network analysis on dependency graphs, while Premraj and Herzig’s replication study [56] demonstrated that simple code and coupling metrics largely matched network measures. SCA cannot see inter-service messaging, broker saturation or cross-host propagation. Architecture recovery tools reconstruct system-level structure from code [57, 58], and HAROS [59] checks ROS systems statically before launch. SaG’s static system analysis (SSA) uses the declared topology instead, propagating code-level metrics across architectural dependencies. Architecture-level coupling metrics count a service’s consumers, as afferent coupling [23], the Absolute Importance of the Service [24] and service fan-in [60]. In publish–subscribe systems these counts are not a single edge lookup, because consumers reach producers only through topics, brokers and shared libraries. They are a typed two-hop query (publisher  $\rightarrow$  topic  $\leftarrow$  subscriber), which SaG’s dependency rules state explicitly; Because this count is exactly the first propagation wave of the reachability oracle used here, this paper reports it as a reference that restates the oracle, not as a predictor (Section 4.4). This lets teams find structural anti-patterns [61, 62] and architectural technical debt [63] in CI/CD [64, 65].

## 2.3. Multi-Criteria Weighting

Aggregating QoS policies or failure effects into one auditable weight is a multi-criteria decision problem, for which the Analytic Hierarchy Process (AHP) [66] is standard; SaG uses it for the topic weights of Section 3.2 and the composite oracle of Section 4.3. AHP’s consistency ratio detects inconsistent judgments but not matrices back-filled from a chosen answer, a distinction this study reports for its weights (Supplementary Section S4).

## 2.4. Graph Learning and Explainability

Centrality indices [21, 22, 67, 68] and cascade models of network robustness [10, 8, 9] assume homogeneous, usually undirected graphs. A single untyped score conflates structurally different elements, such as topics, libraries and hosts, and cannot say *why* a component is critical. Learned node-importance methods, including FINDER [69], DrBC [70] and PowerGraph [71], share the homogeneity assumption, as does learned

approximation of betweenness and closeness [72]. Homogeneous GNNs (GCN [73], GraphSAGE [74], GAT [75]) discard relation identity unless it is supplied as a feature. Softmax attention normalizes over neighbors and so computes a weighted mean that cannot count them, whereas sum aggregation can [76, 77], and counting larger substructures is beyond many message-passing architectures [78]; since the strongest reference here is a neighbor count, this distinction matters (Section 6.2). Heterogeneous GNNs (RGCN [79], HAN [80], HGT [81], MAGNN [82]) learn relation-specific transformations. We evaluate both families under matched capacity (Section 6.2). Khodabandeh et al. [83] apply graph attention to microservice call graphs to predict future interactions. We instead predict the impact of removing a node from a declared topology. GNN explainers such as GNNExplainer [84] and PGExplainer [85] explain models in terms of their internal features. SaG’s proposed explanation layer (Supplementary Section S24) instead names ISO/IEC 25010 quality sub-characteristics [86] and a remediation class for each flagged component; it is not evaluated here.

This study belongs with work showing that simple baselines are often competitive with sophisticated learners [87, 88, 89, 90], and, in software engineering specifically, with the finding that network measures on dependency graphs [55] add little over simple coupling metrics once those are measured [56]. Its contribution is of that kind: a benchmark, with a largely negative result for graph learning, on a pre-deployment ranking task where the simple baseline is a known architecture metric.

### 3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 presents the end-to-end architecture of the SaG framework. The shared front end processes Architecture-as-Code manifests, constructs a typed multigraph, projects implicit runtime interactions throughout a logical dependency layer, and extracts typed node properties. These features feed the ranking engines (Section 4) and, separately and with no shared parameters, the proposed explanation layer (Supplementary Section S24).

#### 3.1. Multigraph Definition

A distributed system is described as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- $V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}}$  holds the five entity types  $\mathcal{T}_V$  of Table 1;  $V_{\text{host}}$  denotes physical or virtual *Execution Hosts*.
- $E$  is the set of directed edges, and  $\tau_V : V \rightarrow \mathcal{T}_V$  and  $\tau_E : E \rightarrow \mathcal{T}_E$  assign entity and relation types.
- $w_V : V \rightarrow (0, 1]$  and  $w_E : E \rightarrow (0, 1]$  weight entity criticality and connection strength. For Applications and Libraries,  $w_V(v) = 1 - \text{CQP}(v)$ , where CQP is a code-quality penalty computed from static code metrics (lines of code, cyclomatic complexity, coupling and cohesion); otherwise  $w_V(v) = 1.0$ .

Table 1: Entity types and structural edge types in the SaG model.

| Entity Type ( $\mathcal{T}_V$ )             | Architectural Role                            | Concrete System Examples                                           |
|---------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------|
| <b>Application</b> ( $V_{\text{app}}$ )     | Process producing/consuming messages          | ROS 2 node, Kafka microservice, MQTT client                        |
| <b>Broker</b> ( $V_{\text{broker}}$ )       | Message routing and queuing intermediary      | RabbitMQ exchange, Mosquitto, EMQX broker                          |
| <b>Topic</b> ( $V_{\text{topic}}$ )         | Named logical communication channel           | <code>/sensor/lidar</code> , <code>orders.payment.completed</code> |
| <b>Execution Host</b> ( $V_{\text{host}}$ ) | Physical or virtualized execution environment | Bare-metal server, Kubernetes worker, Cloud VM                     |
| <b>Library</b> ( $V_{\text{lib}}$ )         | Shared software package or runtime dependency | Kafka client, OpenCV, Protobuf runtime                             |
| Structural Edge ( $\mathcal{T}_E$ )         | Direction                                     | Semantic Meaning                                                   |
| PUBLISHES_TO / SUBSCRIBES_TO                | App/Library $\rightarrow$ Topic               | Component publishes to / consumes from topic                       |
| ROUTES                                      | Broker $\rightarrow$ Topic                    | Broker manages and routes topic traffic                            |
| RUNS_ON                                     | App/Broker $\rightarrow$ Host                 | Process is hosted on host                                          |
| CONNECTS_TO                                 | Host $\rightarrow$ Host                       | Network link between hosts                                         |
| USES                                        | App $\rightarrow$ Library                     | Application links to shared library                                |

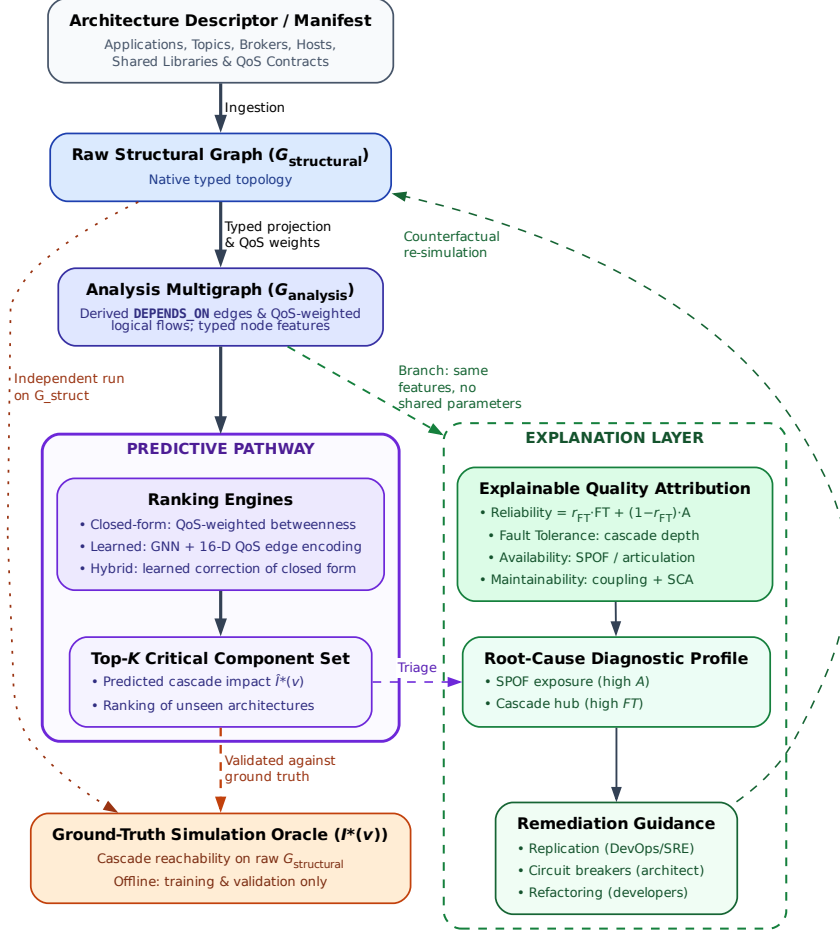


Figure 1: End-to-end architecture of the SaG framework. The predictive pathway runs down the center: manifest ingestion, typed multigraph, **DEPENDS\_ON** projection with typed node properties, the ranking engines (closed-form, learned and hybrid; Figure 3), and the ranked critical set. The dashed edge marks the ground-truth simulation oracles, which operate only on  $G_{\text{structural}}$ , train the predictor offline and take no part in inference. The proposed explanation layer (Supplementary Section S24, not evaluated) reads the same analysis multigraph, shares no parameters with the predictor, and is applied to components after they have been ranked; no output of the predictor flows into it.

Table 2: Notation used throughout the paper.

| Symbol                  | Description                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------|
| $G_{\text{structural}}$ | Raw multigraph (used exclusively by simulation oracles)                                   |
| $G_{\text{analysis}}$   | Logical <b>DEPENDS_ON</b> projection (input to predictors)                                |
| $V_{\text{app}}$        | Application nodes (the primary scored population)                                         |
| $w(t), w(e)$            | QoS topic weight; derived dependency edge weight ( $w(e) = w_E(e)$ on <b>DEPENDS_ON</b> ) |
| $I^*(v)$                | Primary cascade-reachability simulation oracle                                            |
| $I_{\text{dyn}}(v)$     | Dynamic queue-flow discrete-event simulation oracle                                       |
| $I_{\text{comp}}(v)$    | Multi-criteria composite simulation oracle                                                |
| $\rho$                  | Spearman rank correlation coefficient (full population)                                   |
| $\rho_{>0}$             | Spearman rank correlation restricted to active stratum ( $I > 0$ )                        |
| Overlap@K               | Top-K identification set overlap ( $K = 0.20  V_{\text{app}} $ )                          |

### 3.2. Quality-of-Service Link Weighting

A link’s strength depends on its Quality-of-Service (QoS) contract: a **RELIABLE** topic with **TRANSIENT\_LOCAL** durability couples services more strongly than a **BEST\_EFFORT** telemetry stream. Each topic  $t$  carries an aggregate weight  $w(t) \in (0, 1]$  combining declared QoS policies (reliability, durability, priority) with payload size and publication frequency. The sub-weights of reliability, durability and priority come from an Analytic Hierarchy Process (AHP) pairwise-comparison matrix that was stated independently rather than back-solved from a target vector ( $CR = 0.016$ , non-degenerate; Supplementary Section S4, which also shows that three of the framework’s other AHP matrices do encode a declared vector).

On the reachability oracle, QoS weighting does not improve closed-form ranking: unweighted betweenness on the Application–Library projection scores  $\rho = 0.591$ , against 0.553 for QoS-weighted betweenness (**Topo-QoS**; Section 6.1). On the multi-criteria oracle the order reverses, and **Topo-QoS** is the best training-free ranker (Section 6.1). The reference dependency counts reported in this paper are unweighted and do not read  $w(t)$ .

### 3.3. Logical Dependency Projection (**DEPENDS\_ON**)

Structural edges do not directly reflect failure propagation: a subscriber depends on a publisher, yet no direct edge joins them in pub-sub topologies. SaG derives an explicit semantic relation, **DEPENDS\_ON**, directed from *dependent* to *dependency* (“if the target fails, the source is impacted”), via the rules of Table 3.

Table 3: The **DEPENDS\_ON** logical dependency projection rules. <sup>†</sup>Defined for completeness; not exercised by the Application-level evaluation.

| Rule           | Dependency Category     | Structural Pattern (Dependent $\rightarrow$ Dependency)                     | Derived Weight ( $w$ )                                                           |
|----------------|-------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1              | <b>app_to_app</b>       | Subscriber $\rightarrow$ Publisher (via shared Topic)                       | $1 - \prod_{t \in T} (1 - w(t))$                                                 |
| 2 <sup>†</sup> | <b>app_to_broker</b>    | Publisher/Subscriber $\rightarrow$ Broker routing its topics                | $1 - \prod_{t \in T} (1 - w(t))$                                                 |
| 3 <sup>†</sup> | <b>host_to_host</b>     | Host $\rightarrow$ Host (lifted from inter-host app dependencies)           | $\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_E(u \rightarrow v)$ |
| 4 <sup>†</sup> | <b>host_to_broker</b>   | Host $\rightarrow$ Broker (lifted from hosted app dependencies)             | $\max_{u \in \text{hosted}(h)} w_E(u \rightarrow b)$                             |
| 5              | <b>app_to_lib</b>       | Application $\rightarrow$ Shared Library it <b>USES</b>                     | $H(w_V(\text{app}), w_V(\text{lib}))$                                            |
| 6 <sup>†</sup> | <b>broker_to_broker</b> | Broker $\leftrightarrow$ Broker (shared fault-domain colocation, symmetric) | $w_V(\text{host})$                                                               |

Rules 1 and 2 combine topics  $T$  joining a pair by probabilistic union [91, 92, 93], ensuring parallel failure paths increase coupling. Rule 5 uses the harmonic mean  $H(x, y) = 2xy/(x + y)$  [94], and Rules 3 and 4 lift dependencies to hosts by maximum.

**Sequential cascades and simultaneous blasts.** Rule 1 captures sequential cascades, where a failed publisher starves subscribers through queues and buffers. Rule 5 captures simultaneous blasts, where a crashed library takes down all dependent applications at once. Libraries that publish or subscribe are endpoints of Rule 1 in their own right, and an Application reaches a library’s topics transitively through Rule 5. The projection on which every count, **Topo-QoS** and every **-P** learner is computed applies Rule 1 to direct subscriptions only; the repository derivation that produces the node features additionally follows up to three **USES** hops, so an Application subscribing through a library it uses is a Rule 1 dependent there. Rules 2, 3, 4 and 6 represent infrastructural and broker dependencies (Table 3).

**Remark 1 (the dependency count is a typed two-hop count).** Let  $G_{\text{flow}}$  be the Application–Library projection (Rules 1 and 5) and  $v$  an Application. By construction,

$$\text{InDeg}(v) = |\{u \neq v : \exists t \in V_{\text{topic}}, (v, t) \in \text{PUBLISHES\_TO} \wedge (u, t) \in \text{SUBSCRIBES\_TO}\}|. \quad (2)$$

since Rule 5 edges end at Libraries, every edge into an Application is a Rule 1 edge  $u \rightarrow v$ , which exists exactly when  $u \neq v$  (an Application or a Library) directly subscribes to a topic that  $v$  publishes; parallel topics collapse into one edge.

The right-hand side of Eq. (2) is publish–subscribe afferent coupling (fan-in, the Absolute Importance of the Service [23, 24]), and it is a typed two-hop query on the raw multigraph. **InDeg** therefore needs no projection to compute; the projection states which typed query to ask. The equality is also checked on all seventeen corpus graphs (maximum absolute difference 0). What the projection adds beyond this query is measured separately: Rule 5 raises transitive reach by +0.058 (Section 6.1). Because the primary oracle’s first propagation wave is exactly this set (Section 4.4), **InDeg** and its transitive counterpart **Reach** are used in this paper as references that restate the oracle, not as predictors (Amendment 13).

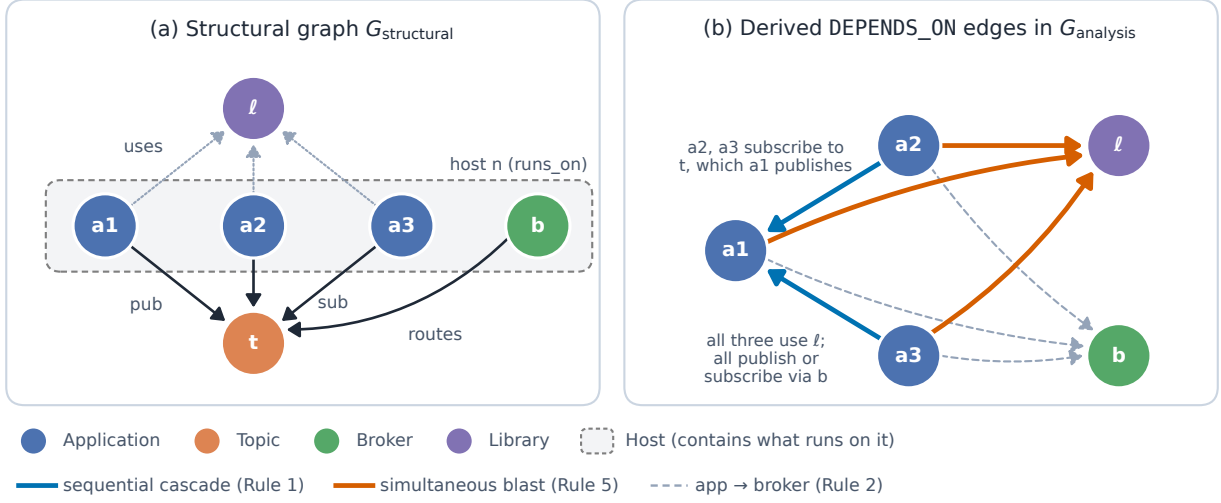


Figure 2: Running example. (a) Three applications share topic  $t$  (routed by broker  $b$ ) and library  $\ell$ , and all run on host  $n$ . No structural edge joins two applications. (b) The derived  $\text{DEPENDS\_ON}$  edges make the hidden dependencies explicit: subscribers  $a_2, a_3$  depend on publisher  $a_1$  (Rule 1), all applications depend on  $\ell$  (Rule 5), and each application depends on the broker (Rule 2). Simulation oracles run on view (a); predictors read view (b).

### 3.4. Dual Graph Views

The **structural graph**  $G_{\text{structural}}$  is the raw deployment topology. The **analysis graph**  $G_{\text{analysis}}$  adds the derived  $\text{DEPENDS\_ON}$  edges and code metrics (Figure 2). Predictor features are computed on  $G_{\text{analysis}}$ , while simulation oracles run strictly on  $G_{\text{structural}}$  (Section 4.4).

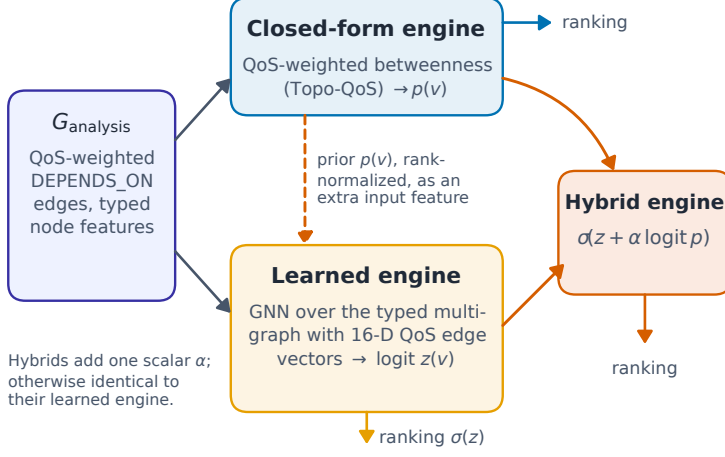
### 3.5. Typed Node Feature Encoding

Both the predictive pathway (Section 4) and the proposed explanation layer (Supplementary Section S24) read typed node properties from  $G_{\text{analysis}}$ . All five entity types share an 18-dimensional base block of normalized topological metrics ( $[0, 1]$ ): PageRank, Reverse PageRank, betweenness, closeness, eigenvector centrality, in- and out-degree, clustering, articulation, bridge ratio, the node QoS weight, incoming and outgoing dependency weights, multi-path coupling, path complexity, fan-out criticality, and the Connectivity Degradation Index (CDI). Four centrality features (PageRank, Reverse PageRank, betweenness and closeness) are computed over weighted edges and so carry QoS information; the nominal QoS-off condition is therefore not completely QoS-free. The in-degree feature, and the incoming dependency weight  $w_{\text{in}}$ , a QoS-weighted version of it, mean that every learner is given a quantity closely related to **InDeg** as an input, so the learners inherit part of the reference counts’ overlap with the oracle through their features (Section 6.2); a learner without both columns was not run. Type-specific blocks extend the vector to 19–25 dimensions (full schema in Supplementary Section S11). CDI evaluates structural connectivity loss via a fixed-size breadth-first sample to control computation time (Section 6.4); its sensitivity to hash-seeded tie breaking is examined in Section 7.3.

## 4. Ranking Engines and Ground Truth

SaG ranks components with three kinds of engine. The **closed-form engine** **Topo-QoS** is QoS-weighted betweenness on the dependency projection (Section 5.2). The **learned engines** are graph neural networks over the typed multigraph and the derived dependency graph (this section). The **hybrid engines** are learned engines that correct the closed-form score (Section 6.1). All are trained or scored against simulation oracles that run on a separate graph view (Sections 4.3–4.4). Figure 3 shows how the three engines relate and how they are evaluated. Full hyperparameters and training commands are on the experiment pages of the replication repository (Section 5.1).

(a) Three ranking engines, one analysis graph



(b) Ground truth and evaluation

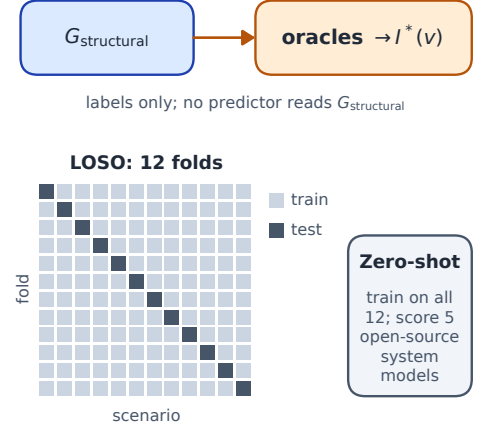


Figure 3: (a) SaG’s three ranking engines read the analysis graph. The closed-form engine scores QoS-weighted betweenness;  $p(v)$  is that **Topo-QoS** score, rank-normalized to  $[0, 1]$  within the graph; the learned engine outputs a logit  $z(v)$ . A hybrid engine gives the learned engine  $p(v)$  as an extra input feature and adds a learned correction to it on the logit scale,  $\sigma(z + \alpha \logit p)$ , with one learnable scalar  $\alpha$ . (b) Ground truth comes from simulation oracles on the structural graph, which no predictor reads. Engines are evaluated by leave-one-scenario-out cross-validation over twelve synthetic architectures and zero-shot on five open-source system models.

#### 4.1. Heterogeneous Graph Transformer and Attention Networks

The primary learned engine HGT-QoS is a three-layer Heterogeneous Graph Transformer (HGT) [81] in PyTorch Geometric [95], with hidden dimension  $D = 64$  and  $H = 4$  heads. Entity-specific linear projections map raw node features  $x_v \in \mathbb{R}^{19-25}$  into hidden space,  $h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v))$ . For each meta-relation  $\langle \tau(u), \phi(e), \tau(v) \rangle$ , attention uses type-parameterized keys, queries, and values, scaled by a learned per-meta-relation prior  $\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}$  (full layer-wise formulations in Supplementary Section S1.1). Message passing runs over both  $G_{\text{analysis}}$  and its transpose. On the raw multigraph, native relations point away from Applications, so forward message passing cannot reach scored nodes, reducing forward GNNs to per-node MLPs over precomputed centralities (Section 7.2). The homogeneous Graph Attention Network baseline (GAT-QoS, and GAT-P-QoS on the dependency projection) uses a 3-layer architecture with 4 attention heads and width  $D = 288$  ( $D = 296$  in the unweighted control to match parameter capacity), projecting all node types into a shared embedding space before homogeneous GATConv layers.

Each directed edge carries a 16-dimensional vector  $e_{uv} \in \mathbb{R}^{16}$ : index 0 represents the coupling weight  $w_E(e)$ , index 1 the normalized path count, indices 2–8 a relation one-hot, and indices 9–15 middleware QoS parameters (reliability, durability, priority, depart-mode flag, deadline pair, max-blocking time). The edge vector is projected and added before attention,  $\tilde{h}_v = h_v + W_{\text{edge}}e_{uv}$ .

#### 4.2. Prediction Head and Training Objective

A composite head predicts simulated cascade impact,  $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{a}_1(v) \parallel \hat{a}_2(v)))$ , where the auxiliary heads  $\hat{a}_1, \hat{a}_2$  are a reliability and a maintainability head.  $\hat{a}_1$  is regressed on the oracle’s reliability column  $I_R^*$ ; the oracle records no maintainability, so  $\hat{a}_2$ ’s regression term is masked out and it is trained only through the composite head. The objective combines regression with listwise and pairwise ranking:

$$\mathcal{L} = \text{MSE}(\hat{I}^*, I^*) + 0.5 \cdot \text{MSE}(\hat{a}_1, I_R^*) + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}}, \quad (3)$$

where  $\mathcal{L}_{\text{rank}}$  is ListMLE [96] over the ground-truth permutation  $\pi$ ,

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left( \hat{s}_{\pi_i} - \log \sum_{j=i}^N \exp(\hat{s}_{\pi_j}) \right), \quad (4)$$

and  $\mathcal{L}_{\text{pairwise}}$  is a margin-ranking loss ( $\gamma = 0.05$ ) over pairs differing by more than  $\gamma$ . Tied labels (about 31% of Applications have  $I^* = 0$ ) enter  $\pi$  in a fixed but arbitrary order, so ListMLE is only approximately defined on them; the pairwise term ignores pairs within  $\gamma$  (Supplementary Section S20).

Models are trained with AdamW ( $\eta = 3 \times 10^{-4}$ , weight decay  $10^{-4}$ ) under cosine warm restarts for up to 300 epochs, with early stopping at patience 60 on an inner validation scenario (the median-sized training scenario, scored whole and never fitted), over five fixed seeds {42, 123, 456, 789, 2024}. Hyperparameters and loss coefficients were fixed before the first sweep and are identical across arms. **This deviates from the registered plan**, whose selection rule chooses every hyperparameter by nested inner-LOSO selection (two-fold inner holdout, Amendment 1); the published learned results use one fixed configuration instead, and no amendment recorded the change (Section 5.3). The registered selection was run afterwards, on the eight-configuration stage-1 grid (depth and rank normalization of features and labels), as a sensitivity analysis (Section 6.2).

#### 4.3. Ground-Truth Simulation Oracles

Ground truth is evaluated using simulation oracles on the raw structural multigraph  $G_{\text{structural}}$ :

- **Primary reachability cascade oracle ( $I^*$ ):** Crashes component  $v$ , propagates outages through dependent topics, brokers and links via breadth-first traversal on  $G_{\text{structural}}$ , and computes the mean fractional feed loss across intact subscribers. Feed loss is scaled by a QoS severity ladder ( $\times 1.2$  RELIABLE,  $\times 1.15$  high priority,  $\times 1.05$  medium) and clamped to  $[0, 1]$ . Each propagation wave is processed in a seeded random order, which decides ties at the propagation threshold; the label is the mean over five seeds. Disabling QoS scaling leaves the Application ordering largely intact ( $\rho = 0.965$  across the twelve folds), and substituting durability-aware rescaling moves it less still ( $\rho = 0.977$ ; Supplementary Section S9), showing that  $I^*$  is predominantly a topological reachability metric.
- **Independent queue-flow discrete-event oracle ( $I_{\text{dyn}}$ ):** A discrete-event SimPy [97] simulation modeling dynamic message emission rates, queue buffer saturation, and network latency. It records the drop in delivered message rate suffered by surviving consumers under active load, and it reads the declared rates, payload sizes and QoS contracts that  $I^*$  ignores. Its mechanism is therefore different from reachability, but its substrate is the same pub-sub topology, and its first-order effect—consumers that lose a failed publisher’s messages—is again subscriber loss. It is labeled for all 1,321 Applications, as the mean over five seeds (Amendment 11), at a cost of 12.7 CPU-hours for the corpus against seconds for  $I^*$ ; an earlier revision used only the first 30 Applications per fold in lexicographic order, which is kept as a sensitivity check (Supplementary Section S36).  $I_{\text{dyn}}$  agrees with  $I^*$  at mean  $\rho = 0.711$  [0.625, 0.794], and its published test–retest reliability is 0.74–0.97 (Supplementary Section S9), which bounds the correlation any deterministic ranker can attain with it.
- **Composite multi-criteria failure oracle ( $I_{\text{comp}}$ ):** The Validate-stage failure simulator, run exhaustively over all 1,321 Applications with QoS weighting on and a primed baseline flow. It scores each removal as  $I_{\text{comp}}(v) = 0.35 \text{RL}(v) + 0.25 \text{FR}(v) + 0.25 \text{TL}(v) + 0.15 \text{FD}(v)$ : reachability loss, fragmentation (0.70 structural component loss + 0.30 severity), throughput loss and flow disruption. The weights come from one of the three AHP matrices that encode a declared vector rather than elicited judgment (Supplementary Section S4), so  $I_{\text{comp}}$  is a stated, not an elicited, notion of harm. Because fragmentation and flow disruption respond to almost every removal,  $I_{\text{comp}}(v) > 0$  for virtually every Application.  $I_{\text{comp}}$  is never a training label.

#### 4.4. Input–Label Independence and Construction Bounds

Features are extracted strictly from  $G_{\text{analysis}}$ , while simulation oracles execute on  $G_{\text{structural}}$ , and no simulation output is exposed as an input feature; a regression test in the replication package enforces this.

Procedural separation does not remove construct overlap, and this is the paper’s central threat, stated once here. **Criterion.** A ranking is a *reference* for an oracle  $O$  if it is a closed-form truncation of  $O$ ’s own computation from the same inputs; it then measures how much of  $O$  is its own rule, not predictive skill, and no contrast against it is a claim. For  $I^*$ , whose first propagation wave is exactly the set **InDeg** counts (Remark 1), the references are its first-order expansion (Eq. (6)), **InDeg**, **Reach** and their raw-graph

equivalents (Amendment 13). For  $I_{\text{dyn}}$  no ranking here truncates its computation, but its first-order effect is the same subscriber loss, so the same rankings are read as references on it too. For  $I_{\text{comp}}$  no ranking is a truncation either; its fragmentation and flow terms, however, respond to removing high-degree and high-betweenness nodes, so **Topo-QoS** and raw degree are *proximate* to it, and their lead on  $I_{\text{comp}}$  is read with the same caution as the references' lead on  $I^*$ . The learned engines are not references by this criterion, but they read in-degree features and are trained on  $I^*$ , so they inherit part of the overlap (Section 6.2 measures how much). Two analyses bound the circularity rather than remove it: every ranker is scored on all three oracles, and its agreement with  $I_{\text{dyn}}$  is also reported after the rank of  $I^*$  (and, separately, of the first-order term) has been partialled out (Section 6.1). Neither replaces validation against observed failures, which this study does not have (Section 7.3).

## 5. Experimental Setup

### 5.1. Corpus and Replication Package

The corpus comprises 2,812 components across seventeen architectures (Table 4). Twelve synthetic topologies form the LOSO folds. They span autonomous vehicles, financial trading, healthcare, industrial SCADA, smart-city IoT, telecom RAN, logistics, gaming, microservices, enterprise integration and air-traffic management. All twelve come from one generator family, as do their code metrics, so LOSO measures transfer across configurations of that generator. Each regenerates byte-identically from a committed configuration, and CI verifies this against a SHA-256 manifest.

Table 4: Evaluation corpus. The twelve synthetic topologies are the LOSO folds; the five open-source system models are excluded from all training and used only for zero-shot transfer (Section 6.3). Counts are read from the committed topology files and verified in CI; per-scenario composition: Supplementary Section S13.

| Dataset                                   | $ V $        | $ V_{\text{app}} $ | Topics     | Brokers   | Hosts      | Libs       | $ E $         |
|-------------------------------------------|--------------|--------------------|------------|-----------|------------|------------|---------------|
| Synthetic evaluation scenarios (11)       | 2,387        | 1,295              | 588        | 60        | 194        | 250        | 10,657        |
| Synthetic case study — ATM (1)            | 74           | 26                 | 27         | 5         | 8          | 8          | 261           |
| <b>Synthetic subtotal (12 LOSO folds)</b> | <b>2,461</b> | <b>1,321</b>       | <b>615</b> | <b>65</b> | <b>202</b> | <b>258</b> | <b>10,918</b> |
| <b>Open-source system models (5)</b>      | <b>351</b>   | <b>141</b>         | <b>120</b> | <b>16</b> | <b>32</b>  | <b>42</b>  | <b>700</b>    |
| <b>Total</b>                              | <b>2,812</b> | <b>1,462</b>       | <b>735</b> | <b>81</b> | <b>234</b> | <b>300</b> | <b>11,618</b> |

**The five open-source systems are hand-authored models.** Autoware.universe (ROS 2), EdgeX Foundry, Home Assistant, and two meshes inspired by Online Boutique and Train-Ticket were each authored by one author as typed multigraphs from public documentation; they are not mechanical extractions. Brokers, QoS profiles, code metrics and host specifications are partly assumed. Two models depart materially from their originals: the Online Boutique model is a 22-application pub-sub mesh with four brokers, whereas the original is an RPC-based service mesh with no broker, and the Train-Ticket model represents its service-discovery server as a broker. Neither contains synchronous call edges. RQ3 therefore tests transfer to independently authored architecture models inspired by open-source systems, not to running deployed systems.

**Replication package.** Datasets, harnesses, checkpoints and result artifacts are archived on Zenodo (see Data Availability). The public repository documents each experiment: its protocol, hyperparameters, `make` target, artifacts and the supplementary section holding its extended results (<https://github.com/onuralpyigit/software-as-a-graph/tree/main/docs/research/jss/experiments>).

### 5.2. Predictors

Table 5 lists the predictors: training-free rankers on the raw multigraph, which need no derivation; training-free rankers on the derived graphs (**Topo**, **Topo-QoS**); learned engines on the raw multigraph and on the dependency graph; and hybrid engines. Dependency counts are not in the table: they restate the primary oracle's propagation rule and are reported as reference rankings (below). On a GNN, **-QoS** means the 16-D QoS edge vector (Section 4.1) together with three QoS node columns ( $w$ ,  $w_{\text{in}}$ ,  $w_{\text{out}}$ ), and **Hybrid-X** is engine X corrected by the closed-form prior. **GAT** and **GAT-QoS** are untyped GATs matched to HGT in parameter

budget, and GAT-QoS reads the same 16-D edge vector as HGT-QoS, so the four learned models form a  $2 \times 2$  over typing (GAT vs. HGT) and the QoS channel. Smaller and projection-based variants that the study also ran are listed in Supplementary Section S30. The learned and hybrid engines ingest the native typed multigraph under LOSO, and the -P learners the Application-Library DEPENDS\_ON graph, with bit-identical node features and labels. In QoS-off arms every edge weight is 1 and the QoS node columns are zeroed, though four centralities still carry QoS (Section 3.5, Supplementary Section S11).

Table 5: Predictors reported in this paper. -QoS: QoS-weighted distances (Topo) or the 16-D QoS edge vector plus three QoS node columns (GNNs); -P: trained on the dependency graph; Hybrid-X: engine X corrected by the Topo-QoS prior. Reference rankings that restate the oracle (Analytic  $I^*$ , InDeg, Reach, Pubs-raw, Reach-R1) and GAT-P+InDeg, which uses one as its prior, are not predictors (Amendment 13). Each dependency-graph learner keeps its raw-multigraph counterpart’s architecture and parameter budget. All runs, with substrate details: Supplementary Section S30.

| Predictor                                                                                | Graph read                         | Typing / edge input                             | Params            |
|------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------|-------------------|
| <i>Training-free, raw multigraph (no derivation; Amendment 12)</i>                       |                                    |                                                 |                   |
| Degree-raw, PR-raw, RevPR-raw                                                            | $G_{\text{structural}}$            | untyped                                         | 0                 |
| <i>Training-free, derived graphs</i>                                                     |                                    |                                                 |                   |
| Topo                                                                                     | application-layer graph            | untyped                                         | 0                 |
| Topo-QoS <sup>¶</sup>                                                                    | App-Lib DEPENDS_ON                 | scalar $w(e)$ ; betweenness only (AP term zero) | 0                 |
| <i>Learned, raw multigraph: typing <math>\times</math> QoS channel at matched budget</i> |                                    |                                                 |                   |
| GAT / GAT-QoS                                                                            | $G_{\text{structural}}$ + features | homogeneous; none / 16-D                        | 437,496 / 429,992 |
| HGT / HGT-QoS                                                                            | $G_{\text{structural}}$ + features | heterogeneous; 1-hot / 16-D                     | 434,620           |
| Hybrid-GAT / Hybrid-HGT                                                                  | $G_{\text{structural}}$ + features | as base + Topo-QoS prior                        | 431,433 / 434,941 |
| <i>Learned, dependency graph (Amendment 9)</i>                                           |                                    |                                                 |                   |
| GAT-P, GAT-P-QoS, HGT-P-QoS                                                              | App-Lib DEPENDS_ON                 | as counterpart                                  | as counterpart    |

**Closed-form scores.** Topo-QoS is computed on the Application-Library DEPENDS\_ON graph (Rules 1 and 5), because on the raw multigraph messages route through topics and brokers and Application betweenness vanishes; the registered comparator Topo reads betweenness from the analysis stage’s application-layer graph:

$$\text{Topo}(v) = 0.6 \cdot \text{BT}(v) + 0.4 \cdot \text{AP}(v), \quad \text{Topo-QoS}(v) = 0.6 \cdot \text{BT}_w(v) + 0.4 \cdot \text{AP}(v), \quad (5)$$

where BT is normalized betweenness,  $\text{BT}_w$  is betweenness over edge distances  $d(e) = 1/(w(e) + 10^{-6})$ , so strongly coupled edges attract shortest paths, and AP flags articulation points. **Defect in the registered comparator.** Because of an implementation defect, the articulation term reads zero for every node, so the reported Topo and Topo-QoS rank exactly as betweenness and QoS-weighted betweenness. They are kept as registered for the confirmatory tests. Restoring the term lowers both (Topo  $0.349 \rightarrow 0.329$ , Topo-QoS  $0.553 \rightarrow 0.533$ ; Supplementary Section S22), so the defect does not favor the reference counts. The two therefore differ in substrate as well as in weighting, and the registered controls of Section 6.1 separate the two.

**Reference rankings.** Five training-free rankings restate the primary oracle’s propagation rule and are reported only as references (Amendment 13): the first-order expansion of  $I^*$  (Analytic  $I^*$ , Eq. (6)); InDeg( $v$ ), the number of direct dependents of  $v$  on the Application-Library DEPENDS\_ON graph, which for an Application equals publish-subscribe afferent coupling and is exactly the set  $I^*$ ’s first wave reaches (Remark 1); Reach( $v$ ), the number of its transitive dependents normalized by  $|V| - 1$ , which is the set a reachability cascade can visit; and their raw-multigraph equivalents Pubs-raw, the number of topics a component publishes, and Reach-R1, the transitive closure of publisher  $\rightarrow$  topic  $\rightarrow$  subscriber paths (Reach without Rule 5). A reference shows how much of an oracle a restatement of its rule recovers, which is the size of the circularity, not predictive skill. References carry no contrast against Topo-QoS, and a learner’s distance from them is reported as such, not as a test.

**Raw-multigraph baselines.** Three rankers are computed on  $G_{\text{structural}}$  without any derived edge (Amendment 12): total degree (Degree-raw) and unweighted PageRank ( $\alpha = 0.85$ ) on the raw graph and on its reverse (PR-raw, RevPR-raw).

### 5.3. Metrics, Protocols and Statistics

**Population.** Every predictor is scored on the Application set  $V_{\text{app}}$ .

**Metrics.** Ranking is measured by Spearman  $\rho$  against the simulation oracles. To separate rank ordering among active components from the trivial agreement on inert components with zero impact, we report both full-population Spearman  $\rho$  and active-stratum Spearman  $\rho_{>0}$  (restricting to components with true impact  $> 0$ ). Critical-set identification is measured by  $\text{Overlap}@K$ , the fraction of the true top- $K$  recovered by the predicted top- $K$  with  $K = \text{round}(0.20 |V_{\text{app}}|)$ . PR-AUC,  $F_1@ \tau$  and nDCG@10 are reported in Supplementary Section S15.

**Protocols.** Under *LOSO*, learned models train on eleven scenarios and are tested on the twelfth, over all 12 folds and five seeds; the reported value is the mean over seeds of each seed’s  $\rho$ . Under *zero-shot transfer*, models trained on all twelve scenarios are evaluated without fine-tuning on the five system models. Training-free rankers are fitted to nothing, so for them both protocols reduce to *per-scenario evaluation* on the same twelve scenarios and five models; they are placed in the same tables only so that every row is scored on identical scenarios and labels.

**Statistics.** Summary  $\rho$  is the arithmetic mean of the per-fold (or per-system) Spearman correlations, with a 95% CI from a bootstrap that resamples folds ( $B = 2,000$ ) [98]; comparisons use paired two-sided Wilcoxon signed-rank tests over folds [99]. With 12 folds, the smallest attainable two-sided  $p$  is 0.00049. Fisher- $z$  averaging leaves the order of every row of Table 6 unchanged. Weighting folds by  $|V_{\text{app}}|$  leaves the top five unchanged (the reference **InDeg** rises to 0.810) but lifts **Topo-QoS** to 0.596, level with the raw-multigraph learned engines (0.595–0.606) (Supplementary Section S35). Spearman  $\rho$  assigns tied scores their average rank.  $\text{Overlap}@K$  breaks ties at the  $K$  boundary by node order, which is arbitrary for integer counts; Section 6.1 therefore also reports tie-aware recall curves. Because LOSO folds share ten of eleven training scenarios, every registered contrast is also tested with the Nadeau–Bengio corrected resampled  $t$  [100] and, under  $|V_{\text{app}}|$  weighting, with an exact sign-flip permutation test (Supplementary Section S36).

**Analysis plan and the status of each result.** The analysis plan and its amendments are a version-controlled file in the replication repository, each entry committed before the results it governs, with dates and commit hashes listed in Supplementary Section S25; they are not deposited with an independent registry, so their timestamps are the repository’s own. One rule assigns every result to one of three tiers. *Confirmatory*: only the original plan (2026-09-06), **HGT-QoS** vs. **Topo-QoS** (primary) and **HGT** vs. **Topo-QoS**, Holm-corrected across the two. *Registered secondary*: every amendment written before any result of its own arms existed—the matched  $2 \times 2$  (Amendment 2), the two hybrids (Amendments 5 and 6, 2026-09-23/24), the reference counts and QoS controls (Amendment 7), the dependency-graph learners (Amendment 9), the value of the derivation (Amendment 10), the full-population  $I_{\text{dyn}}$  and learned combinations (Amendment 11) and the round-8 arms (Amendment 14), each Holm-corrected within its own family. All of them were written after the primary null was known, and the choice of arms was informed by it. *Exploratory*: analyses registered after their results existed (Amendments 8, 12 where marked, and 13) and every descriptive analysis. Amendment 13 reclassifies the dependency counts as references (Section 4.4); it changes no number. **One deviation from the plan**: its selection rule chooses every hyperparameter by nested inner selection, but the published learned results use one fixed configuration, and no amendment recorded that change; Amendment 14 runs the registered selection as a sensitivity analysis (Section 6.2). As a sensitivity check, the 13 contrasts of the plan and Amendments 2, 5 and 6 are also pooled under one omnibus Holm correction (`reproduce/omnibus_holm.py`): both hybrid contrasts against **Topo-QoS** stay below 0.05 (Hybrid-GAT  $p_{\text{omni}} = 0.019$ , Hybrid-HGT 0.041), and the primary contrast does not ( $+0.069$ , nominal  $p = 0.266$ ,  $p_{\text{omni}} \geq 0.46$ ).

## 6. Results

All results are reported on the Application population ( $V_{\text{app}}$ ) against the primary oracle  $I^*(v)$ , and, for the rankers of Table 7, also against the queue-flow oracle  $I_{\text{dyn}}(v)$  and the multi-criteria oracle  $I_{\text{comp}}(v)$  (Section 4.3). Everything except the primary contrast, the matched  $2 \times 2$  and the two hybrid contrasts is exploratory (Section 5.3). Per-fold results, secondary strata and extended protocol notes are in the Supplementary Material and public experiment pages (Section 5.1). Figure 4 summarizes the main findings.

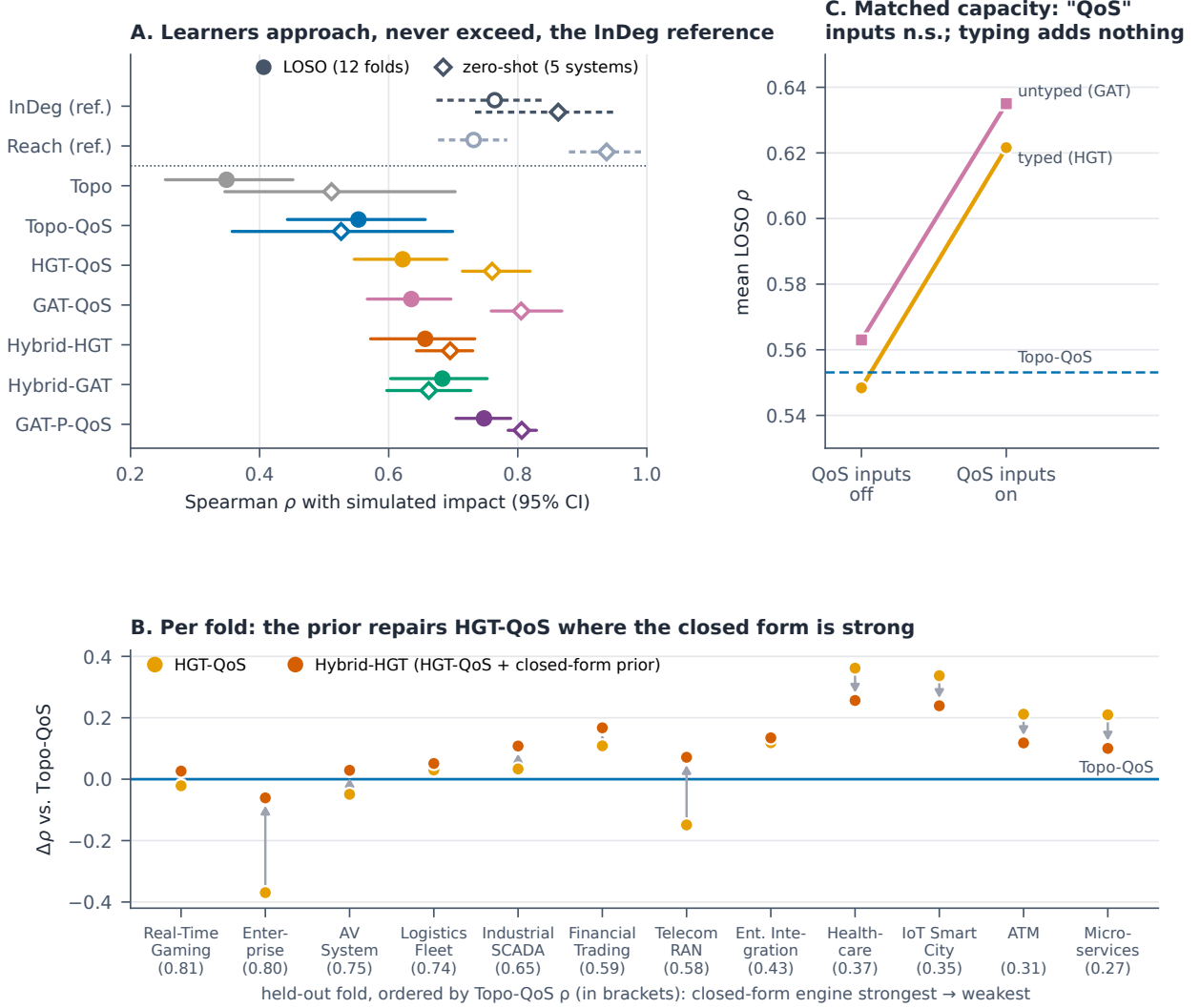


Figure 4: Main results at a glance, Application population. (A) Mean Spearman  $\rho$  with 95% bootstrap intervals under LOSO (filled circles; Table 6) and zero-shot on five system models (open diamonds). Grey hollow markers are reference rankings that restate  $I^*$ 's propagation rule (**InDeg**, **Reach**; Amendment 13), not predictors; the neural model reading the dependency graph (**GAT-P-QoS**) approaches **InDeg** without exceeding it. (B) Per held-out fold, the gain of **HGT-QoS** and of **Hybrid-HGT** over **Topo-QoS**. (C) Cell means of the capacity- and channel-matched  $2 \times 2$  (Table 8): the "QoS" inputs, which include a QoS-weighted in-degree column, raise both models by about 0.07 (not significant after Holm), while typed and untyped configurations are indistinguishable.

### 6.1. RQ1: Ranking Accuracy Against Structural Baselines

*Summary.* Against  $I^*$ , the strongest rankings are references that restate its first propagation wave (first-order expansion 0.808, direct dependents 0.764). The registered primary contrast, HGT-QoS against Topo-QoS (0.553), is null. The two hybrids beat Topo-QoS on 11 of 12 folds (Holm  $p = 0.0068$  and  $0.0029$ ), but not their own base learners, and only Hybrid-GAT beats unweighted betweenness after correction. The best GNN is a GAT on the dependency graph (0.748); it is statistically indistinguishable from the direct-dependent count, but not equivalent to it within  $\pm 0.05$ , and it depends on its degree features (Section 6.2). On full-population  $I_{dyn}$  no GNN reaches the references, while a gradient-boosted surrogate trained on  $I_{dyn}$  exceeds them; on  $I_{comp}$  closed-form centrality and raw degree, which are proximate to its terms, rank highest. Recovering 80% of the true top-20% set requires flagging the top 40% by GAT-P-QoS.

Learned engines are trained on eleven scenarios and scored on the twelfth; training-free rankers are simply scored on each of the twelve (26 to 300 Applications,  $K$  between 5 and 60). Table 6 reports all of them against  $I^*$ , with the reference rankings of Section 5.2 in a separate block: they restate  $I^*$ 's rule, so their rows measure the circularity and carry no contrast.

Table 6: Main results under LOSO across twelve synthetic architectures (Application population; learned engines: five seeds, CPU sweeps).  $\Delta\rho$  is paired by fold against Topo-QoS, with a bootstrap 95% CI ( $B = 2,000$ ) and a two-sided Wilcoxon signed-rank test.  $\rho_{>0}$  denotes Spearman correlation restricted to active components ( $I^* > 0$ ).  $p_{Holm}$  is given within each registered family: hybrids (Amendments 5 and 6, confirmatory). The reference block holds rankings that restate  $I^*$ 's propagation rule (the post hoc first-order expansion, Eq. (6); the dependency counts of Amendment 7; their raw-graph equivalents of Amendment 12); they are not predictors and carry no contrast (Amendment 13). <sup>‡</sup>Exploratory: Amendment 9 registered its contrasts against the references and raw counterparts. <sup>§</sup>Raw-multigraph rankers (Amendment 12, exploratory; unadjusted  $p$ ); PR-raw is constant for every Application and has no defined  $\rho$ . <sup>¶</sup>Topo and Topo-QoS carry an implementation defect (articulation term zero; Section 5.2); corrected values are 0.329 and 0.533. Bold: best predictor per column. Per-fold values, contrasts among references, and GAT-P+InDeg, whose prior is a reference: Supplementary Sections S23, S33, S34 and S35.

| Predictor                                                                                            | LOSO $\rho$ [95% CI]        | Active $\rho_{>0}$ | $\Delta\rho$ vs Topo-QoS [95% CI] | Won   | $p$ ( $p_{Holm}$ )  | Overlap@K    |
|------------------------------------------------------------------------------------------------------|-----------------------------|--------------------|-----------------------------------|-------|---------------------|--------------|
| <i>Reference: restatements of <math>I^*</math>'s propagation rule (not predictors; Amendment 13)</i> |                             |                    |                                   |       |                     |              |
| Analytic $I^*$                                                                                       | 0.808 [0.755, 0.853]        | 0.631              | —                                 | —     | —                   | 0.536        |
| InDeg (direct dependents)                                                                            | 0.764 [0.674, 0.840]        | 0.516              | —                                 | —     | —                   | 0.504        |
| Reach (transitive dependents)                                                                        | 0.732 [0.674, 0.782]        | 0.286              | —                                 | —     | —                   | 0.344        |
| Pubs-raw <sup>§</sup>                                                                                | 0.731 [0.637, 0.810]        | 0.410              | —                                 | —     | —                   | 0.487        |
| Reach-R1 <sup>§</sup>                                                                                | 0.674 [0.606, 0.738]        | 0.088              | —                                 | —     | —                   | 0.300        |
| <i>Training-free, raw multigraph (no derivation)<sup>§</sup></i>                                     |                             |                    |                                   |       |                     |              |
| Degree-raw                                                                                           | 0.199 [0.068, 0.328]        | 0.218              | −0.354 [−0.475, −0.236]           | 1/12  | 0.0015              | 0.307        |
| RevPR-raw                                                                                            | 0.089 [−0.019, 0.192]       | 0.075              | −0.465 [−0.558, −0.364]           | 0/12  | 0.0005              | 0.199        |
| <i>Training-free, application layer</i>                                                              |                             |                    |                                   |       |                     |              |
| Topo <sup>¶</sup>                                                                                    | 0.349 [0.254, 0.452]        | 0.174              | −0.204 [−0.286, −0.122]           | 0/12  | 0.0005              | 0.366        |
| <i>Training-free, dependency graph</i>                                                               |                             |                    |                                   |       |                     |              |
| Topo-QoS <sup>¶</sup>                                                                                | 0.553 [0.443, 0.657]        | 0.280              | —                                 | —     | —                   | 0.388        |
| <i>Learned and hybrid, on the raw multigraph</i>                                                     |                             |                    |                                   |       |                     |              |
| HGT-QoS                                                                                              | 0.622 [0.547, 0.690]        | 0.312              | +0.069 [−0.046, +0.174]           | 8/12  | 0.266               | 0.426        |
| GAT-QoS                                                                                              | 0.635 [0.567, 0.696]        | 0.338              | +0.082 [−0.046, +0.201]           | 7/12  | 0.233               | 0.438        |
| Hybrid-HGT                                                                                           | 0.657 [0.572, 0.733]        | 0.345              | +0.103 [+0.055, +0.152]           | 11/12 | 0.0034 (0.0068)     | 0.435        |
| Hybrid-GAT                                                                                           | 0.683 [0.603, 0.753]        | 0.362              | +0.130 [+0.075, +0.190]           | 11/12 | 0.0015 (0.0029)     | 0.450        |
| <i>Learned, on the dependency graph (Amendment 9)</i>                                                |                             |                    |                                   |       |                     |              |
| GAT-P-QoS                                                                                            | <b>0.748</b> [0.704, 0.789] | <b>0.440</b>       | + <b>0.195</b> [+0.100, +0.294]   | 10/12 | 0.0068 <sup>‡</sup> | <b>0.454</b> |
| HGT-P-QoS                                                                                            | 0.514 [0.392, 0.636]        | 0.237              | −0.039 [−0.155, +0.077]           | 4/12  | 0.622 <sup>‡</sup>  | 0.380        |

**The reference level.** By Remark 1, InDeg is publish–subscribe afferent coupling [23, 24, 60] and counts exactly the Applications that  $I^*$ 's first propagation wave reaches; **Reach** counts the transitive dependents a reachability cascade can visit. Their values on  $I^*$  (0.764,  $\rho_{>0} = 0.516$ ; 0.732) therefore measure how much of the oracle a restatement of its rule recovers, not predictive skill, and the same holds for their raw-graph equivalents: counting the topics a component publishes (Pubs-raw) reaches 0.731, and the raw publisher  $\rightarrow$  topic  $\rightarrow$  subscriber closure (Reach-R1) 0.674. Among the references, the derivation contributes in one measured place: without Rule 5, **Reach** falls by 0.058 (9/12 folds, Holm  $p = 0.0068$ ; Amendment 10). Untyped raw-graph metrics recover none of the reference: total degree reaches 0.199 and reverse PageRank 0.089, and PageRank is constant for every Application because Applications have no incoming raw edges.

**Learners on the dependency graph approach the reference.** The same graph neural networks, trained on the dependency graph instead of the raw multigraph, reach  $\rho = 0.748$  (GAT-P-QoS), the best GNN in Table 6, and 0.016 below the direct-dependent count on the per-seed statistic; its five-seed ensemble is 0.007 above it (0.772, Table 7). Neither difference is significant, and neither is equivalence: the 90% interval of the per-fold difference reaches  $\pm 0.076$ , so a two one-sided test at the registered margin of  $\pm 0.05$  fails ( $p = 0.17$  per seed, 0.13 for the ensemble; Amendment 14, F3). The learner and the count are therefore indistinguishable at this sample size, not ranked. The approach to the reference rests on the degree features every learner receives (Section 3.5): removing `in_degree` and  $w_{\text{in}}$  lowers GAT-P-QoS to 0.612 (Section 6.2). A variant that uses `InDeg` as its prior is reported only in Supplementary Section S34, because its prior is a reference. HGT-P-QoS reaches only 0.514 and is highly seed-dependent (mean within-fold seed SD 0.254; Section 7.3).

**Hybrids outperform the registered comparator, and little else.** Hybrid-HGT reaches  $\rho = 0.657$  (+0.103, Holm  $p = 0.0068$ ) and Hybrid-GAT 0.683 (+0.130, Holm  $p = 0.0029$ ) against Topo-QoS, each on 11 of 12 folds; both stay significant under the omnibus Holm correction ( $p_{\text{omni}} = 0.041$  and 0.019), under the Nadeau–Bengio corrected  $t$  ( $p = 0.019$  and 0.014) and under  $|V_{\text{app}}|$ -weighted averaging (exact sign-flip  $p = 0.038$  and 0.005; Supplementary Section S36). They are the study’s only significant registered improvements, and they are weaker than they look, because a hybrid receives  $p(v)$  as an input and adds a learned correction to its logit, so it nests its comparator. Against their own base learners, which Amendments 5 and 6 registered alongside, neither is significant (Hybrid-HGT vs. HGT-QoS +0.035,  $p = 0.73$ ; Hybrid-GAT vs. GAT-QoS +0.048,  $p = 0.30$ ). Against the two stronger training-free controls of Amendment 7 (Amendment 14, F7), Hybrid-GAT keeps a margin (unweighted betweenness on the same graph +0.092, constant topic weights +0.088, both Holm  $p = 0.049$ ), Hybrid-HGT does not (+0.065 and +0.061, Holm  $p = 0.085$ ). The prior also costs transfer: zero-shot, both hybrids fall below their base learners (Section 6.3). On their own, the raw-multigraph learned engines do not differ significantly from Topo-QoS (HGT-QoS +0.069,  $p = 0.266$ , the null primary contrast; GAT-QoS +0.082,  $p = 0.233$ ).

**Where the closed-form gain comes from.** Topo reads betweenness from the application-layer graph, whereas Topo-QoS computes QoS-weighted betweenness on the Application–Library dependency graph. Three controls registered in Amendment 7 separate substrate from weighting: unweighted betweenness on the same graph scores 0.591; constant topic weights score 0.595; permuting QoS profiles scores 0.559 ( $-0.006$ ,  $p = 0.73$ ). On  $I^*$ , the gain comes from the Application–Library graph structure, not from QoS contract content.

**The first-order expansion of the oracle.** The strongest reference is a post hoc closed-form first-order expansion of  $I^*$ :

$$\hat{I}_1^*(v) = \sum_{t \in \text{pub}(v)} \frac{|\text{sub}(t)|}{|\text{pub}(t)|}, \quad (6)$$

where  $|\text{pub}(t)| \geq 1$  holds for all published topics  $t \in \text{pub}(v)$  by construction ( $v$  publishes to  $t$ ), preventing any division by zero. It reaches mean  $\rho = 0.808$  [0.755, 0.853] ( $\rho_{>0} = 0.631$ ): direct subscriber loss is most of what  $I^*$  measures, and `InDeg` is a coarser version of the same quantity. As a reference it bounds the mean rather than every fold: on four folds (AV, Financial Trading, Healthcare, Industrial SCADA) `InDeg` slightly exceeds it (by 0.005–0.038).

**Beyond the reachability oracle.** Table 7 scores the rankers against the two other oracles. On full-population  $I_{\text{dyn}}$ , which agrees with  $I^*$  at  $\rho = 0.711$  [0.625, 0.794], the references stay ahead of every GNN (first-order expansion 0.706, `InDeg` 0.664, against 0.496–0.615), and Topo-QoS reaches 0.471. Because  $I_{\text{dyn}}$  shares the topology and the first-order effect with  $I^*$ , the partial correlation shows how much of each ranking survives outside the oracle it was fitted to or restates. After the rank of  $I^*$  is removed, `InDeg` keeps 0.272 [0.188, 0.366], `Reach` 0.117, Topo-QoS 0.134 and the GNNs 0.107–0.173, every interval excluding zero. After the first-order expansion is removed instead, the order reverses: `InDeg` keeps 0.079, and the raw-multigraph GNNs keep most (0.212–0.217), so the learners carry queue-flow signal beyond the direct count that the count does not. Learning from the right labels matters more than the graph model. A gradient-boosted model over the counts, the first-order expansion and QoS features, trained on  $I_{\text{dyn}}$  labels of the training scenarios, ranks held-out architectures at 0.799, above the first-order expansion (+0.094, Holm  $p = 0.019$ ) and `InDeg` (+0.135, Holm  $p = 0.019$ ; Amendment 11, Family B); its QoS features carry that gain (+0.095

Table 7: Rankers against three oracles, twelve LOSO folds, Application population (registered secondary and exploratory). All three oracles are exhaustive over the 1,321 Applications;  $I_{\text{dyn}}$  is the five-seed mean of Amendment 11 (the earlier lexicographic  $n = 30$  sample is in Supplementary Section S36). Partial  $\rho$ : Spearman correlation with  $I_{\text{dyn}}$  after the rank of  $I^*$ , or of the first-order expansion  $\hat{I}_1^*$  (Eq. (6)), is regressed out of both. Published learned rows score the mean of the five seeds’ *predictions*, which raises  $\rho$  on  $I^*$  above the per-seed means of Table 6; Amendment 11’s GBM rows and Amendment 14’s surrogate are per-seed means. \*Trained on  $I_{\text{dyn}}$  labels of the eleven training scenarios: a surrogate for  $I_{\text{dyn}}$ , not an independent check of it. Reference rows restate  $I^*$ ’s rule (Section 4.4). Italics: highest value per column among predictors. 95% CIs bootstrap folds.

| Ranker                                                                                                                        | $I^* \rho$ | $I_{\text{dyn}} \rho$ [95% CI] | Partial $\rho(\cdot, I_{\text{dyn}}   I^*)$ [95% CI] | Partial $\rho(\cdot, \hat{I}_1^*   I^*)$ [95% CI] | $I_{\text{comp}} \rho$ |
|-------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------|------------------------------------------------------|---------------------------------------------------|------------------------|
| <i>Reference: restatements of <math>I^*</math>’s propagation rule (not predictors)</i>                                        |            |                                |                                                      |                                                   |                        |
| Analytic $I^*$                                                                                                                | 0.808      | 0.706 [0.629, 0.783]           | 0.318 [0.234, 0.412]                                 | —                                                 | 0.636                  |
| InDeg                                                                                                                         | 0.764      | 0.664 [0.562, 0.758]           | 0.272 [0.188, 0.366]                                 | 0.079                                             | 0.650                  |
| Reach                                                                                                                         | 0.732      | 0.583 [0.510, 0.651]           | 0.117 [0.055, 0.184]                                 | 0.102                                             | 0.302                  |
| Pubs-raw                                                                                                                      | 0.731      | 0.645 [0.543, 0.737]           | 0.255 [0.179, 0.332]                                 | 0.056                                             | 0.616                  |
| <i>Training-free predictors (proximate to <math>I_{\text{comp}}</math>; Section 4.4)</i>                                      |            |                                |                                                      |                                                   |                        |
| Topo-QoS                                                                                                                      | 0.553      | 0.471 [0.381, 0.559]           | 0.134 [0.061, 0.204]                                 | −0.091                                            | 0.702                  |
| Degree-raw                                                                                                                    | 0.199      | 0.232 [0.134, 0.329]           | 0.149 [0.089, 0.227]                                 | −0.091                                            | 0.719                  |
| <i>Learned GNNs (seed ensemble)</i>                                                                                           |            |                                |                                                      |                                                   |                        |
| GAT-P-QoS                                                                                                                     | 0.772      | 0.615 [0.549, 0.673]           | 0.111 [0.016, 0.213]                                 | 0.182                                             | 0.274                  |
| Hybrid-GAT                                                                                                                    | 0.702      | 0.599 [0.514, 0.669]           | 0.173 [0.082, 0.267]                                 | 0.056                                             | 0.585                  |
| Hybrid-HGT                                                                                                                    | 0.672      | 0.573 [0.497, 0.638]           | 0.162 [0.074, 0.252]                                 | 0.029                                             | 0.582                  |
| HGT-QoS                                                                                                                       | 0.667      | 0.549 [0.464, 0.622]           | 0.130 [0.035, 0.224]                                 | 0.212                                             | 0.201                  |
| GAT-QoS                                                                                                                       | 0.647      | 0.523 [0.436, 0.606]           | 0.107 [0.018, 0.199]                                 | 0.217                                             | 0.144                  |
| HGT-P-QoS                                                                                                                     | 0.618      | 0.496 [0.395, 0.601]           | 0.112 [0.033, 0.190]                                 | 0.087                                             | 0.334                  |
| <i>Learned combinations of dependency signals (Amendment 11; GBM over counts, Analytic <math>I^*</math> and QoS features)</i> |            |                                |                                                      |                                                   |                        |
| GBM-Dep-QoS                                                                                                                   | 0.818      | 0.718                          | —                                                    | —                                                 | 0.483                  |
| GBM-Dep-QoS→dyn*                                                                                                              | 0.764      | 0.799                          | —                                                    | —                                                 | 0.477                  |
| GAT-P-QoS→dyn* (Amendment 14)                                                                                                 | 0.734      | 0.598                          | —                                                    | —                                                 | 0.368                  |

over the same model without them, Holm  $p = 0.014$ ), whereas on  $I^*$  they add nothing ( $-0.003$ ,  $p = 0.62$ ; Family C). The GNN given the same labels does not: **GAT-P-QoS**→dyn reaches 0.598, below the first-order expansion ( $-0.108$ , Holm  $p = 0.014$ ) and the gradient-boosted surrogate ( $-0.201$ , Holm  $p = 0.0015$ ), and no better than the same GNN trained on  $I^*$  ( $-0.008$ ; Amendment 14, F5). These surrogates are trained on the oracle they are scored on; their value is cost, since labeling  $I_{\text{dyn}}$  took 12.7 CPU-hours for the corpus (Section 6.4). Trained on  $I^*$  instead, the same gradient-boosted model is less robust across the three oracles than the first-order expansion alone (worst-case  $\rho -0.121$ , Holm  $p = 0.001$ ; Family A).

**The multi-criteria oracle favors different rankers.** On  $I_{\text{comp}}$ , which adds fragmentation, throughput and flow disruption (Section 4.3), QoS-weighted centrality ranks above the direct-dependent reference ( $\rho = 0.702$  [0.649, 0.753] against 0.650; the reference ahead on only 3 of 12 folds,  $p = 0.110$ ), and raw total degree, the weakest ranker on  $I^*$ , reaches 0.719 (the reference ahead on 4 of 12 folds, Holm  $p = 0.220$ ), while **Reach** falls to 0.302. Both leaders are proximate to  $I_{\text{comp}}$ ’s fragmentation and flow terms (Section 4.4), so their lead is read as the same kind of overlap as the references’ lead on  $I^*$ , not as predictive skill. The learned engines without a centrality prior collapse on this oracle (0.144–0.334; the sum-aggregation GNNs of Section 6.2 reach 0.41–0.44); the two hybrids, which carry **Topo-QoS** as a prior, keep 0.58. No ranker is best on all three oracles.

**Identifying the critical set.** Rank correlation over all Applications is not the operational question in a CI gate, which is whether the true top of the ranking gets flagged. Figure 5 gives, for each ranker, the share of the true top-20% set on  $I^*$  that falls within its top  $k\%$ , with ties resolved in expectation (Amendment 12). At  $k = 20\%$ , **GAT-P-QoS** recovers 0.48 of the critical set and **Topo-QoS** 0.38, consistent with their  $\text{Overlap}@K$  (0.454 and 0.388): about half of what a top-20% gate flags is not in the true top 20%, and about half of the true top 20% is missed. Recovering 80% of the critical set requires flagging the top 40% by **GAT-P-QoS** (0.81), and **Topo-QoS** does not reach 80% within 50%. The references do little better: at  $k = 20\%$ , **InDeg** recovers 0.49 (tie-breaking bounds 0.48–0.50), and 80% needs the top 45% by **InDeg** (0.83) or by the first-order expansion (0.89); only the first-order expansion reaches 90% within the top 50%, and **Reach** does not reach 80% within 50%. On the earlier  $n = 30$   $I_{\text{dyn}}$  sample (the right panel of Figure 5; not recomputed on the full

population), GAT-P-QoS reaches 0.65 at 40% and 0.80 at 50%, against 0.83 at 40% for the InDeg reference.

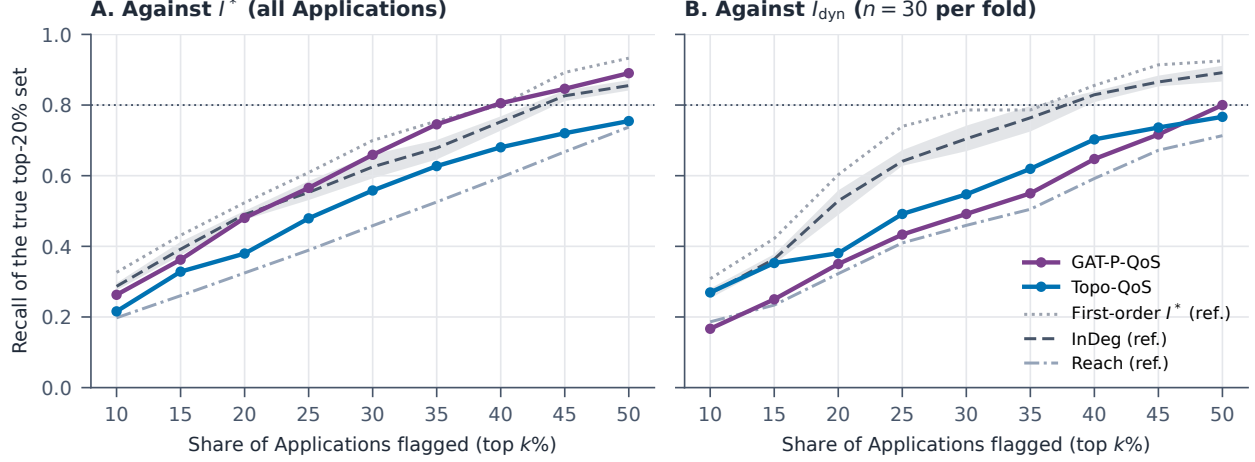


Figure 5: Recall of the true top-20% set (tie-inclusive) by each ranker’s top  $k\%$ , mean over the twelve LOSO folds, against  $I^*$  (left) and the  $n = 30$   $I_{\text{dyn}}$  sample (right). Ties are resolved in expectation over random tie-breaking; grey dashed curves are reference rankings that restate  $I^*$ ’s rule (Amendment 13), and the shaded band gives InDeg’s optimistic and pessimistic bounds. Reach, whose scores tie heavily, has much wider bounds (0.12–0.80 at  $k = 20\%$  on  $I^*$ ), so its expected curve is the only meaningful summary. The dotted line marks 80% recall.

## 6.2. RQ2: What Learned Engines Need

*Summary.* Learned engines gain +0.08 to +0.11 from reading the dependency graph, where messages reach the scored Applications from their dependents (11–12 of 12 folds). What brings the dependency-graph GAT close to the reference is its degree features: without them it loses 0.14 (Holm  $p = 0.005$ ). A sum-aggregation GNN, which can count dependents, loses only 0.01 without them, although the aggregator contrast itself is not significant after correction. With capacity matched, relation typing adds nothing; with the weighted in-degree held in both arms, the “QoS” inputs add nothing measurable either (+0.030, Holm  $p = 0.33$ ).

Table 8: The  $2 \times 2$  with capacity and edge-channel width matched (Amendment 2): GAT (–T–Q, 437,496 parameters), HGT (T–Q, 434,620), GAT-QoS (–TQ, 429,992), HGT-QoS (TQ, 434,620). One CPU sweep, twelve LOSO folds, five seeds, Application population. Holm correction across the three orthogonal quantities. Cell means: 0.563, 0.548, 0.635, 0.622.

| Quantity                                                              | Contrast                  | $\Delta\rho$ | 95% CI           | Won   | $W$  | $p$   | $p_{\text{Holm}}$ |
|-----------------------------------------------------------------------|---------------------------|--------------|------------------|-------|------|-------|-------------------|
| <i>Three orthogonal quantities, Holm-corrected across these three</i> |                           |              |                  |       |      |       |                   |
| Typing (main effect)                                                  | averaged over Q           | −0.014       | [−0.052, +0.023] | 4/12  | 29.0 | 0.470 | 0.940             |
| “QoS” inputs (main effect)                                            | averaged over T           | +0.073       | [+0.013, +0.120] | 10/12 | 13.0 | 0.043 | 0.127             |
| Typing × “QoS” interaction                                            | difference of differences | +0.001       | [−0.050, +0.042] | 6/12  | 34.0 | 0.733 | 0.940             |
| <i>Simple effects — descriptive, not separately corrected</i>         |                           |              |                  |       |      |       |                   |
| Typing, QoS absent                                                    | HGT vs. GAT               | −0.015       | [−0.064, +0.033] | 5/12  | 31.0 | 0.569 | —                 |
| Typing, QoS present                                                   | HGT-QoS vs. GAT-QoS       | −0.013       | [−0.054, +0.026] | 4/12  | 27.0 | 0.380 | —                 |
| “QoS” inputs, typing absent                                           | GAT-QoS vs. GAT           | +0.072       | [+0.028, +0.109] | 10/12 | 9.0  | 0.016 | —                 |
| “QoS” inputs, typing present                                          | HGT-QoS vs. HGT           | +0.073       | [−0.002, +0.136] | 10/12 | 19.0 | 0.129 | —                 |

**The “QoS” factor was mostly the weighted in-degree.** In Table 8 its main effect (+0.073) is not significant after Holm correction ( $p_{\text{Holm}} = 0.127$ ), and it could not be attributed to QoS content in any case:  $w_{\text{in}}$ , the sum of incoming dependency weights, is a QoS-weighted version of InDeg, and the QoS-off arms zeroed it. Amendment 14 reran the matched  $2 \times 2$  with  $w_{\text{in}}$  kept in the QoS-off cells (Table 9, F4): the main effect falls to +0.030 [−0.023, +0.085] (Holm  $p = 0.33$ ), typing is −0.036 and the interaction +0.046, none significant. More than half of the earlier QoS effect was therefore the weighted in-degree. The closed-form controls point the same way: unweighted betweenness (0.591) and constant topic weights (0.595) match or exceed QoS-weighted betweenness (0.553), and permuting QoS profiles across topics changes nothing

Table 9: Amendment 14 arms: twelve LOSO folds, five seeds, one CPU invocation with every comparator re-run (each reproduces its published per-seed  $\rho$  exactly), Application population.  $\Delta\rho$  is paired by fold against the comparator named, with a bootstrap 95% CI; Holm within each registered family (F1: degree features; F2: aggregator; F4: the  $2 \times 2$  with  $w_{\text{in}}$  held).  $-\text{deg}$ : `in_degree` and  $w_{\text{in}}$  zeroed;  $-\text{deg}^*$ : also PageRank, closeness and eigenvector centrality, whose correlation with `InDeg` is 0.74–0.77 on average (Supplementary Section S36). Zero-shot: mean  $\rho$  on the five system models. Registered secondary.

| Arm                                                                                                                                                                                              | Comparator                | LOSO $\rho$ ( $I^*$ ) | $\Delta\rho$ [95% CI]   | Won  | $p_{\text{Holm}}$ | $I_{\text{dyn}}$ / $I_{\text{comp}}$ | Zero-shot |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------|-------------------------|------|-------------------|--------------------------------------|-----------|
| <i>F1: removing the degree features</i>                                                                                                                                                          |                           |                       |                         |      |                   |                                      |           |
| GAT-P-QoS–deg                                                                                                                                                                                    | GAT-P-QoS (0.748)         | 0.612                 | −0.136 [−0.205, −0.072] | 1/12 | 0.0049            | 0.485 / 0.156                        | 0.821     |
| GAT-P-QoS–deg*                                                                                                                                                                                   | GAT-P-QoS                 | 0.587                 | −0.161 [−0.252, −0.080] | 1/12 | 0.0049            | 0.462 / 0.138                        | 0.654     |
| GAT-QoS–deg                                                                                                                                                                                      | GAT-QoS (0.635)           | 0.365                 | −0.270 [−0.344, −0.199] | 0/12 | 0.0015            | 0.295 / −0.047                       | 0.810     |
| <i>F2: sum aggregation (GINE, 434,123 parameters) instead of softmax attention</i>                                                                                                               |                           |                       |                         |      |                   |                                      |           |
| GIN-P-QoS                                                                                                                                                                                        | GAT-P-QoS                 | 0.732                 | −0.016 [−0.055, +0.017] | 7/12 | 0.910             | 0.606 / 0.407                        | 0.806     |
| GIN-P-QoS–deg                                                                                                                                                                                    | GAT-P-QoS–deg             | 0.721                 | +0.108 [+0.027, +0.197] | 8/12 | 0.157             | 0.602 / 0.428                        | 0.810     |
| GIN-P-QoS–deg*                                                                                                                                                                                   | GAT-P-QoS–deg*            | 0.711                 | +0.124 [+0.031, +0.230] | 9/12 | 0.157             | 0.589 / 0.438                        | 0.776     |
| <i>F4: matched <math>2 \times 2</math> with <math>w_{\text{in}}</math> held (cells GAT+<math>w_{\text{in}}</math> 0.628, HGT+<math>w_{\text{in}}</math> 0.569, GAT-QoS 0.635, HGT-QoS 0.622)</i> |                           |                       |                         |      |                   |                                      |           |
| Typing (main effect)                                                                                                                                                                             | averaged over Q           | —                     | −0.036 [−0.079, +0.004] | 2/12 | 0.330             | —                                    | —         |
| “QoS” inputs (main effect)                                                                                                                                                                       | averaged over T           | —                     | +0.030 [−0.023, +0.085] | 9/12 | 0.330             | —                                    | —         |
| Interaction                                                                                                                                                                                      | difference of differences | —                     | +0.046 [−0.009, +0.101] | 9/12 | 0.330             | —                                    | —         |

( $\Delta = -0.006$  [−0.025, +0.015],  $p = 0.733$ ). On  $I^*$  we find no evidence that QoS contract content carries ranking signal; on  $I_{\text{dyn}}$ , whose mechanics read the contracts, it does (Section 6.1, Amendment 11 Family C). **What the dependency-graph learner needs.** Every learner receives in-degree and  $w_{\text{in}}$  as node features, and on the dependency graph it also aggregates exactly the dependent neighborhood, so its closeness to `InDeg` could come from either. Amendment 14 separates the two (Table 9). Removing the two degree columns costs GAT-P-QoS 0.136, and removing the three centralities most correlated with in-degree as well costs 0.161 (F1, rule F1a): the approach to the reference depends on the degree features. That is what message-passing theory predicts for this architecture: softmax attention normalizes over neighbors and computes a weighted mean, which cannot count them, whereas a sum aggregator can [76, 77]. A GINE network of the same parameter budget bears this out descriptively, keeping 0.721 and 0.711 without the same columns, 0.01–0.02 below its own full-feature score, but the aggregator contrasts miss significance after Holm correction (+0.108 and +0.124,  $p_{\text{Holm}} = 0.157$ ; rule F2b), so the study does not claim an aggregator effect. No dependency-graph learner, with or without degree features, is equivalent to `InDeg` within  $\pm 0.05$  (equivalence bounds 0.063–0.276; F3). Zero-shot, the degree columns matter little (0.810–0.821 without them), because on the system models most of the ranking is separating inert from active components (Section 6.3).

**Relation typing under matched capacity.** HGT-QoS remains within 0.013 of GAT-QoS (Table 8). On the raw multigraph this says little about typing, because no message reaches an Application there: a gradient-boosted model over the same per-node features without any graph (GBM-Feat, Amendment 8) reaches 0.642, level with GAT-QoS, and removing HGT’s reverse pass, the only route by which messages reach Applications, changes nothing (HGT-QoS-U, −0.010,  $p = 0.91$ ; Supplementary Section S25). On the dependency graph, where messages do arrive, HGT-P-QoS reaches 0.514 with a mean within-fold seed SD of 0.254, against 0.748 and 0.030 for GAT-P-QoS. This is one fixed configuration (width 100), and averaging its five seeds’ predictions raises it to 0.618; we therefore read it as seed instability of that configuration, not as evidence against relational typing in general.

### 6.3. RQ3: Zero-Shot Transfer to Models of Open-Source Systems

*Summary.* On five hand-authored models of open-source systems, the best learned engines rank  $I^*$  at about 0.81 zero-shot (GAT-P-QoS 0.806, GAT-QoS 0.805), far above closed-form centrality (0.511–0.526), but their active-stratum agreement is weak ( $\rho_{>0} \leq 0.342$ ). The reference rankings reach higher (transitive dependents 0.938,  $\rho_{>0} = 0.871$ ; direct dependents 0.863), because on these models most of  $I^*$  is reachability itself. The models differ structurally from the synthetic corpus, and all five were written by one author, so these numbers describe transfer to those models, not to deployed systems. The centrality prior costs transfer: both hybrids fall below their base learners (0.695 and 0.662 against 0.760 and 0.805).

The **Reach** reference reaches  $\rho = 0.938$  and  $\rho_{>0} = 0.871$  on the five models: 0.836–0.997 on the three originally publish–subscribe systems and 0.966–0.998 on the two RPC systems re-expressed as pub-sub meshes (per system: Supplementary Table S36). These values exceed anything **Reach** reaches on the synthetic

Table 10: Zero-shot transfer to hand-authored models inspired by five open-source systems (Application population). Models are evaluated out-of-distribution without fine-tuning.  $\rho_{>0}$  denotes rank correlation restricted to active components ( $I^* > 0$ ). PR-AUC measures critical-set identification quality. All rows are scored on the same labels (the learned engines’ zero-shot label files; Amendment 12, R5). The training-free rows are not trained on anything, so “zero-shot” applies to the learned rows only. Reference rows restate  $I^*$ ’s propagation rule and are not predictors (Amendment 13). Bold: best predictor per column. 95% CIs bootstrap the five systems and are indicative only.

| Predictor                                                                              | Evaluation Substrate | Mean $\rho$ [95% CI]        | Active $\rho_{>0}$ | PR-AUC       |
|----------------------------------------------------------------------------------------|----------------------|-----------------------------|--------------------|--------------|
| <i>Reference: restatements of <math>I^*</math>’s propagation rule (not predictors)</i> |                      |                             |                    |              |
| Reach                                                                                  | Dependency graph     | 0.938 [0.879, 0.991]        | 0.871              | 0.933        |
| InDeg                                                                                  | Dependency graph     | 0.863 [0.734, 0.952]        | 0.321              | 0.752        |
| <i>Training-free predictors</i>                                                        |                      |                             |                    |              |
| Topo                                                                                   | Application layer    | 0.511 [0.346, 0.703]        | −0.104             | 0.474        |
| Topo-QoS                                                                               | Dependency graph     | 0.526 [0.357, 0.699]        | −0.088             | 0.474        |
| <i>Learned and hybrid, raw multigraph</i>                                              |                      |                             |                    |              |
| HGT-QoS                                                                                | Raw multigraph       | 0.760 [0.714, 0.819]        | 0.236              | 0.713        |
| GAT-QoS                                                                                | Raw multigraph       | 0.805 [0.759, 0.868]        | 0.319              | 0.790        |
| Hybrid-HGT                                                                             | Raw multigraph       | 0.695 [0.643, 0.730]        | 0.210              | 0.602        |
| Hybrid-GAT                                                                             | Raw multigraph       | 0.662 [0.597, 0.727]        | 0.185              | 0.600        |
| <i>Learned, dependency graph</i>                                                       |                      |                             |                    |              |
| GAT-P-QoS                                                                              | Dependency graph     | <b>0.806</b> [0.785, 0.829] | <b>0.342</b>       | <b>0.838</b> |

folds (0.732,  $\rho_{>0} = 0.286$ ), and the difference is structural. Half of the system models’ Applications have zero simulated impact (0.51 on average, against 0.31 on the folds), and their fan-in is more concentrated (Gini 0.65 against 0.50; Supplementary Section S35). On such graphs, separating the inert half from the rest already produces a high full-population  $\rho$ , and **Reach**, which is zero exactly for components with no dependents, does that perfectly. The active-stratum  $\rho_{>0}$  is the more informative column here. On it every predictor is weak (learned engines 0.185–0.342, closed-form centrality negative), as is the direct-dependent reference (0.321 [−0.044, 0.687]), while **Reach** (0.871) is far ahead: on these models, ordering active components by transitive reach is close to what  $I^*$  computes, which is the circularity at its strongest, not transferable skill.

#### 6.4. RQ4: Analysis Cost

*Summary.* The reference counting path (projection plus **InDeg**), which approximates  $I^*$ ’s first wave, takes 1–29 ms per corpus architecture and 1.1 s at 10,000 components; one labeling pass of the reachability oracle takes 0.08–4.5 s on the corpus and grows roughly quadratically, to 347 s at 2,000 components and 36 minutes at 5,000. The neural pipeline is dominated by feature extraction, which costs median  $5.6\times$  a direct run of the oracle; inference itself takes 56 ms at 2,000 nodes. No energy was measured.

Table 11: Per-stage latency of the inference pipeline across graph sizes (CPU, median of 3 runs; 5 for the forward pass). The analysis stage is stable across repeats (p10–p90 within 1% of the median), while the forward pass is dominated by interpreter and dispatch overhead; the 249-node row carries first-call warm-up.

| V     | E      | Analyze | Graph →    | HGT          | Analyze :     | Forward      |
|-------|--------|---------|------------|--------------|---------------|--------------|
|       |        | (s)     | Tensor (s) | Forward (ms) | Forward       | p10–p90 (ms) |
| 249   | 1,127  | 1.74    | 0.010      | 26.5         | 66×           | 13.0–34.4    |
| 499   | 2,402  | 8.32    | 0.022      | 16.4         | 509×          | 15.6–16.4    |
| 999   | 6,422  | 44.54   | 0.056      | 21.1         | 2,108×        | 19.1–36.5    |
| 1,998 | 19,301 | 239.34  | 0.157      | 56.2         | <b>4,259×</b> | 43.8–57.8    |

Table 11 breaks down the neural pipeline. Its analysis stage, which computes the node features including betweenness and a directed articulation score, dominates; across the corpus it takes 2.0–17.7 $\times$  (median 5.6 $\times$ ) the time of running  $I^*$  directly, so where the  $I^*$  ranking itself is wanted, running the oracle is cheaper than

Table 12: The counting path against one labeling pass of the reachability oracle  $I^*$  on generated graphs of increasing size (Amendment 12, R6; CPU, median of 5 runs;  $I^*$  timed once at 4,995 components and not at 9,990, where one pass would take hours). Projection: deriving the Application–Library `DEPENDS_ON` graph from the manifest.

| $ V $ | $ E $   | Projection +<br>InDeg (ms) | Reach<br>(ms) | $I^*$ pass<br>(s) | $I^*$ :<br>count |
|-------|---------|----------------------------|---------------|-------------------|------------------|
| 249   | 1,081   | 8.0                        | 11.7          | 4.0               | 500×             |
| 499   | 2,437   | 11.2                       | 24.9          | 15.1              | 1,351×           |
| 999   | 6,372   | 25.3                       | 113.1         | 58.5              | 2,313×           |
| 1,998 | 19,242  | 109.7                      | 490.8         | 346.6             | 3,159×           |
| 4,995 | 94,790  | 426.8                      | 4,369.9       | 2,163.5           | 5,069×           |
| 9,990 | 348,277 | 1,095.3                    | 23,302.4      | not timed         | —                |

approximating it with a learned engine. Table 12 shows the counting path. On the corpus it takes 1–29 ms per architecture against 0.08–4.5 s for one  $I^*$  pass, and on generated graphs the gap widens with size: the count grows roughly with the number of edges,  $I^*$  roughly with the square of the number of components. At 5,000 components one  $I^*$  pass takes 36 minutes and the count 0.43 s. Because the count restates  $I^*$ ’s first wave, this is the cost of approximating the oracle, not of an independent prediction: the count’s practical advantage is cost at scale and in inner loops, not accuracy over the oracle it approximates. **Reach** is costlier than **InDeg** (23 s at 10,000 components) because it computes a transitive closure. These are wall-clock measurements on one CPU; no energy was measured, and we make no sustainability claim beyond them.

## 7. Discussion, Threats to Validity, and Limitations

### 7.1. Practical Consequences

**Two regimes.** The value of a ranker depends on what the oracle it approximates costs. *Where the oracle is cheap*, as the reachability oracle  $I^*$  is (0.05–4.5 s per corpus architecture; Section 6.4), a practitioner who wants its ranking can compute it directly, from the same manifest and with no operational profile. The benchmark’s plainest finding belongs here: under a reachability notion of impact, publish–subscribe afferent coupling—a direct-dependent count that architecture research has used for two decades [23, 24]—ranks as well as every graph neural network evaluated, at two to three orders of magnitude lower cost than the oracle. It does so because it restates the oracle’s first wave (Section 4.4), which is also why it is not credited here with predictive skill; its practical role is as a cheap stand-in for  $I^*$  in inner loops and on large architectures. *Where the oracle is expensive*, as the queue-flow simulator  $I_{\text{dyn}}$  is (12.7 CPU-hours to label the corpus), a learned surrogate earns its place: a gradient-boosted model over the dependency counts, the first-order expansion and QoS features, trained on  $I_{\text{dyn}}$  labels of other architectures, ranks held-out architectures at  $\rho = 0.799$ , above every closed-form approximation (Section 6.1). The GNN trained on the same labels does not (0.598). Nothing in this study shows that either oracle, or any proxy of them, predicts observed outages.

**Choosing a ranker.** Table 13 summarizes the evidence by oracle for predictors only. Where the reachability ranking itself is wanted, running  $I^*$  is the simplest option at corpus size. Among predictors on  $I^*$ , the dependency-graph GAT (GAT-P-QoS, 0.748) and the hybrids (0.657–0.683) rank best, all need training, and none is distinguishable from the direct-dependent count. Where the queue-flow notion of impact matters, a surrogate trained on queue-flow labels is the only ranker that beats the closed-form approximations, and the only place where declared QoS contracts carry measurable signal. Where failure is understood as fragmentation and throughput loss ( $I_{\text{comp}}$ ), QoS-weighted centrality or raw total degree ranks best (0.702 and 0.719), with the caveat that both are proximate to that oracle’s terms. On unfamiliar topologies like the five system models, the plain learned engines transfer at about 0.81 against 0.51–0.53 for centrality, but their active-stratum agreement is weak and the result rests on five models by one author.

**How much to flag.** A top-20% gate on GAT-P-QoS recovers about half of the true top-20% set on  $I^*$  (0.48), and the rest of what it flags is not in that set. The recall curves of Figure 5 give the trade-off directly: 80% of the critical set is recovered by flagging the top 40% by GAT-P-QoS (0.81), Topo-QoS does not reach it within the top 50%, and 90% is not reached by any predictor within the top 50%. The references do no better (InDeg needs the top 45%). A gate that must catch most critical components therefore has to flag

nearly half the system, which limits any of these rankers as a filter; their value is as an ordering for review, not as a pass/fail threshold.

Table 13: Which predictor the evidence supports, by the failure notion the oracle encodes (registered secondary and exploratory; Section 6). Reference rankings are excluded because they restate  $I^*$ ’s propagation rule (Section 4.4).

| Failure notion (oracle)                            | Ranker                                                                | Evidence                                                                                                                                                   |
|----------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reachability cascade ( $I^*$ )                     | Run $I^*$ directly                                                    | 0.05–4.5 s per corpus architecture; best predictor <b>GAT-P-QoS</b> $\rho = 0.748$ , indistinguishable from, not equivalent to, the direct-dependent count |
| Queue-flow delivery loss ( $I_{\text{dyn}}$ )      | Surrogate trained on $I_{\text{dyn}}$ (GBM-Dep-QoS $\rightarrow$ dyn) | 0.799 vs. first-order expansion 0.706 (Holm $p = 0.019$ ); costs 12.7 CPU-hours of labels; the GNN surrogate reaches 0.598                                 |
| Fragmentation and throughput ( $I_{\text{comp}}$ ) | <b>Topo-QoS</b> or Degree-raw                                         | $\rho = 0.702 / 0.719$ , both proximate to the oracle’s terms; learned engines $\leq 0.585$                                                                |
| Unfamiliar topology                                | <b>GAT-P-QoS</b> or <b>GAT-QoS</b>                                    | $\rho = 0.806 / 0.805$ vs. <b>Topo-QoS</b> 0.526; $\rho_{>0} \leq 0.342$ ; five single-author models                                                       |

## 7.2. When Graph Learning Helps, and When It Does Not

Practitioners also need to understand the boundary conditions under which learned models add value, whether relation-specific typing is warranted, and where learning fails. Table 14 summarizes the empirical evidence across accuracy terciles and architectural paradigms (extended per-fold details in Supplementary Section S32).

Table 14: Where graph learning helps. LOSO folds are grouped into terciles of closed-form accuracy (**Topo-QoS**  $\rho$ ); system models are grouped by architectural paradigm. Mean Spearman  $\rho$  against  $I^*(v)$ , Application population;  $\rho_{>0}$ : active stratum. For system models, “learned” ranges over the pure learned engines and “closed-form” over **Topo** and **Topo-QoS**. The last column gives the reference counts (Amendment 7), which restate  $I^*$ ’s rule and are not predictors (Amendment 13), and the dependency-graph learner (Amendment 9). “Best choice” ranges over predictors only. Exploratory. Because the terciles are formed on the comparator’s own score, a learner’s gain over **Topo-QoS** is inflated in the lowest tercile and deflated in the highest by regression to the mean; with four folds per tercile the pattern is descriptive only.

| Regime                                | Where                                                    | Best choice              | Evidence: raw-multigraph engines                                                                                                     | Dependency graph (references; learner)                          |
|---------------------------------------|----------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Closed-form ranks poorly              | Microservices, ATM, IoT Smart City, Healthcare           | Dependency-graph learner | <b>Topo-QoS</b> 0.324; <b>HGT-QoS</b> 0.604, <b>GAT-QoS</b> 0.626, both 4/4 folds. The prior trims the gain (hybrids 0.502 / 0.554). | <b>InDeg</b> 0.693, <b>Reach</b> 0.700, <b>GAT-P-QoS</b> 0.716. |
| Intermediate                          | ESB, Telecom RAN, Financial Trading, Industrial SCADA    | Dependency-graph learner | <b>Topo-QoS</b> 0.561; <b>HGT-QoS</b> 0.589, <b>GAT-QoS</b> 0.660; Hybrid-HGT 0.681, Hybrid-GAT 0.697, both 4/4 folds.               | <b>InDeg</b> 0.742, <b>GAT-P-QoS</b> 0.760.                     |
| Closed-form ranks well                | Logistics Fleet, AV System, Enterprise, Real-Time Gaming | Hybrid engine            | <b>Topo-QoS</b> 0.775; <b>HGT-QoS</b> 0.672 (1/4 folds), <b>GAT-QoS</b> 0.620 (0/4); hybrids 0.786 / 0.798.                          | <b>InDeg</b> 0.858, <b>GAT-P-QoS</b> 0.768.                     |
| Unlike the corpus, originally pub-sub | Autoware, EdgeX, Home Assistant                          | Learned engine           | Learned 0.716–0.927 vs. closed-form 0.289–0.534; learned $\rho_{>0}$ 0.183–0.833 where closed-form is negative.                      | <b>Reach</b> 0.836–0.997; $\rho_{>0}$ 0.674–0.971.              |
| Unlike the corpus, originally RPC     | Online Boutique, Train-Ticket models                     | Mixed                    | Learned 0.710–0.810, but <b>Topo</b> 0.891 on Online Boutique; learned $\rho_{>0}$ $-0.19$ to $+0.16$ .                              | <b>Reach</b> 0.966–0.998; $\rho_{>0}$ 0.813–0.976.              |

**On the raw multigraph, learning operated primarily over node features.** Three controls clarify how the raw-multigraph engines functioned. First, gradient-boosted decision trees over per-node features without graph message passing (**GBM-Feat**) achieve  $\rho = 0.642$  under LOSO, level with **GAT-QoS** (0.635). Second, on the raw multigraph, all structural relations point away from Applications, meaning standard forward GAT convolutions deliver no messages to the scored nodes: deleting every edge leaves their outputs unchanged.

Third, HGT reaches Applications only through reverse edge passes, and ablating them incurs no measurable loss (HGT-QoS-U,  $-0.010$ ,  $p = 0.91$ ; Supplementary Section S25). Thus, on the raw multigraph, neural models acted as learned non-linear combiners of precomputed centralities and degree metrics rather than relational message-passing engines. On the derived dependency graph, however, directed edges flow directly from dependents to dependencies: 3-layer message passing aggregates the true dependent neighborhood, enabling GAT-P-QoS to outperform GBM-Feat by  $+0.106$  (Supplementary Section S34). Because every learner also receives an in-degree feature (Section 3.5), this margin is not evidence that attention discovers the dependent count unaided, and Amendment 14 shows it does not: without the degree columns GAT-P-QoS loses  $0.136$ , whereas a sum-aggregation network of the same size loses  $0.011$  (Section 6.2).

**Homogeneous versus heterogeneous architectures.** On the raw multigraph, where message passing into Applications was inactive, typing added nothing measurable (main effect  $-0.014$ , interaction  $+0.001$ ). On the dependency graph, where message passing operates, the heterogeneous transformer (HGT-P-QoS) was unstable across seeds in the single configuration evaluated (width 100, fixed configuration; mean within-fold seed standard deviation  $0.254$ , mean  $\rho = 0.514$ ), whereas the homogeneous GAT-P-QoS was not ( $0.030$ ,  $\rho = 0.748$ ). This supports a narrow statement: *in this configuration*, relation-specific parameters bought no accuracy and cost stability. It does not show that typing is harmful in general. Only HGT-QoS was run under the registered selection rule (Section 6.2), averaging five seeds’ predictions lifts HGT-P-QoS to  $0.618$  (still below the references), and denser topologies or architectures whose messages reach scored nodes may behave differently.

### 7.3. Threats to Validity

**Construct validity and oracle circularity.** All ground truth is simulated, and no ranker is validated against observed failures. The circularity between the primary oracle and the rankers is analyzed once, in Section 4.4: the dependency counts are references for  $I^*$ , closed-form centrality is proximate to  $I_{\text{comp}}$ , and the learned engines inherit part of the overlap through their degree features, which Amendment 14 shows they depend on. Inert components are separated by construction, which is why  $\rho_{>0}$  is reported beside  $\rho$ . The reclassification of the counts (Amendment 13) was decided after all results existed and changed no number. The two other oracles bound, but do not remove, the threat:  $I_{\text{dyn}}$  shares the topology and the first-order effect, and  $I_{\text{comp}}$ ’s weights encode a declared rather than an elicited priority vector. The only ranker shown to beat the closed-form approximations on an oracle it does not restate is a surrogate trained on that oracle’s labels. The evaluation is confined to Applications, so dependency Rules 2–4 and 6, which govern brokers and hosts, are untested.

**Internal validity, reproducibility, and drift.** Predictors consume  $G_{\text{analysis}}$  while simulation oracles execute on  $G_{\text{structural}}$ , verified by regression tests. Model capacity, depth and early stopping were held constant across comparison arms. The registered selection rule was not applied to the published learned results, a deviation disclosed in Section 5.3; Amendment 14 applies it to HGT-QoS and GAT-P-QoS as a sensitivity check (Section 6.2), and every other statement about a learned engine is conditional on one fixed configuration. The learned engines are also seed-sensitive. Across the five seeds of a fold, the mean within-fold standard deviation of  $\rho$  is  $0.099$  for HGT-QoS,  $0.254$  for HGT-P-QoS,  $0.024$  for GAT-QoS and  $0.030$  for GAT-P-QoS, and single HGT seeds differ from others in the same fold by more than  $1.0$  (Supplementary Section S35). Averaging the five seeds’ *predictions* instead of their  $\rho$  values raises every learned engine, most of all the unstable ones (HGT-P-QoS from  $0.514$  to  $0.618$ ; Supplementary Section S35); the tables report the registered statistic, the mean of per-seed  $\rho$ . Across revision cycles and compute devices, learned cells drifted by up to  $0.172$  ( $\pm 0.041$  on HGT-QoS), more than several effects this paper reports (the primary contrast,  $+0.069$ ; the QoS main effect,  $+0.073$ ). For that reason every learned cell in Tables 6, 8 and 10 is reported from one named CPU sweep whose per-seed logs reproduce the published mean exactly, the Amendment 14 sweep re-ran every learned comparator to an exact match, and a hierarchical bootstrap that resamples seeds within folds widens the learned rows’ intervals only slightly (Supplementary Section S36). The Connectivity Degradation Index samples nodes with hash-dependent tie-breaking, so PYTHONHASHSEED=0 is fixed. All reported values are reconciled mechanically against released artifacts.

**External validity and the single-modeller threat.** Synthetic topologies come from a single generator family, and the five system models are hand-authored approximations (22–41 Applications) of open-source

projects, each written by the first author from public documentation. All five RQ3 results are exposed to this threat, and it is not mitigated: no model was authored independently by a second modeler, and no inter-modeller agreement is reported. The system models also differ structurally from the synthetic folds (Supplementary Section S35): a larger share of their Applications has zero simulated impact and their fan-in is more concentrated, which favors **Reach** and inflates full-population  $\rho$ . Two of the five are RPC systems re-expressed as publish–subscribe meshes. Architectures with tens of thousands of components were not evaluated; the counting path was timed up to 10,000 components (Section 6.4), the learned engines and the oracles were not. Published learned criticality models (FINDER, DrBC) were not reproduced, because they do not support typed multigraphs or pub-sub semantics.

**Conclusion validity.** The only confirmatory results are the plan’s two contrasts, the null primary and the null secondary (Section 5.3). Everything else, including the two hybrid contrasts, is registered secondary or exploratory and was designed after the primary null was known. LOSO folds share ten of eleven training scenarios, so the Wilcoxon  $p$ -values are nominal and anti-conservative; the Nadeau–Bengio corrected test, which accounts for the overlap, leaves the hybrid contrasts significant and the primary null (Supplementary Section S36). For training-free rankers the folds are simply twelve scenarios from one generator, so the effective sample is small, and with twelve folds a bootstrap CI over folds is coarse; over five system models it is indicative only. Equivalence, which a claim that one ranker “matches” another requires, is not established for any pair at  $\pm 0.05$ .

#### 7.4. Limitations and Future Work

The study leaves open, and states as limitations: validation against observed outages or post-mortems, for which benchmark systems with cataloged faults, such as Train-Ticket [29], and the MQTT-based EdgeX and Home Assistant deployments are the natural first targets; a full hyperparameter search for every learned arm, of which only a stage-1 selection for two engines was run; an independently authored second model of each open-source system, with inter-modeller agreement; automatic extraction of architecture graphs from ROS 2 launch files, Kubernetes manifests and Helm charts; energy measurement; synchronous call edges, which the dependency rules do not model; and the proposed explanation layer (Supplementary Section S24), whose attributions have not been validated in any way. A natural next step is a GNN surrogate for the queue-flow oracle that uses sum aggregation, since the gradient-boosted surrogate shows the labels carry learnable signal that the attention-based GNN did not recover.

## 8. Conclusion

Software-as-a-Graph (SaG) derives explicit dependency graphs from publish–subscribe deployment manifests and ranks components by simulated cascade impact before deployment. We used it to benchmark closed-form centrality, graph neural networks and hybrid engines on twelve synthetic architectures and five hand-authored models of open-source systems, against three simulators.

The registered primary contrast, a heterogeneous graph transformer against weighted centrality, was null; the only confirmatory gains are two hybrids that correct a learned engine with the centrality prior and beat centrality on 11 of 12 synthetic architectures. The reachability simulator’s first propagation wave is exactly a component’s number of direct dependents, so dependency counts were reported as references that restate the oracle, not as predictors: the oracle’s first-order expansion reached  $\rho = 0.808$  and the direct-dependent count 0.764. Graph attention networks reading the dependency graph approached that reference (0.748) but never exceeded it, and because they read an in-degree feature and are trained on the oracle, they do not escape the circularity. On a queue-flow simulator no predictor reached the reference; on a multi-criteria simulator, weighted centrality ranked above every learned engine. No predictor identifies the critical set sharply: catching 80% of the true top fifth requires flagging the top 40% even for the best learned engine.

For practice, the evidence supports a modest recommendation. Where the reachability ranking is wanted and the architecture is of the size studied here, running the simulator directly is cheaper than approximating it with a learned engine, and learned engines add nothing over a restatement of its rule; where failure means fragmentation and throughput loss, weighted centrality is the better choice. A ranker that can be credited

with predictive skill on this task must be scored against an oracle it does not restate, and whether any of these rankings, or the simulators themselves, predict real outages remains the open question.

## Declarations

**CRediT Authorship Contribution Statement.** **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision. **Declaration of Competing Interest.** The authors declare no competing financial interests or personal relationships that could have influenced this work. **Funding.** This research received no external grant. **Data Availability.** The replication package (datasets, harnesses, checkpoints, scripts) is available on Zenodo under DOI 10.5281/zenodo.14922108 [101] with `uv/pip` environments. The public repository documents every experiment reported here — protocol, hyperparameters, reproduction command, artifacts and extended results — at <https://github.com/onuralpyigit/software-as-a-graph/tree/main/docs/research/jss/experiments>. Synthetic datasets regenerate byte-identically. The deposit ships all artifacts backing reported tables as a dated bundle (`SaG_JSS_Results_<stamp>`) with a `MANIFEST.json` recording SHA-256 digests, commit hashes, and corpus provenance. Four supplementary artifacts predate provenance stamping and carry no commit or corpus digest: `atm_scale_sweep_v3.json` (S6), `qos_label_ablation.json` (Section 4.3), `threshold_sensitivity_v3.json` (S3) and `topic_weight_sensitivity_v3.json` (S1); their correspondence to the corpus is asserted by the bundle rather than recorded in the file. In the main text, only the two label-ablation correlations of Section 4.3 ( $\rho = 0.965$  and  $0.977$ ) rest on one of them; the other three back supplementary sensitivity sweeps only, and no number in the abstract or in Tables 6–10 depends on any of the four. The verification script (`reproduce/reconcile_manuscript.py`) runs standalone against the deposit, mechanically verifying 1,347 reported figures in the manuscript and supplement against the JSON artifacts.

## Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic’s Claude to assist with typesetting, LaTeX formatting, the development of analysis and reporting scripts in the replication package, and drafting revisions of the manuscript text in response to review comments. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The study design, the choice of experiments, the interpretation of results, and all scientific claims are the authors’ own.

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